

24 Aug 2026 17:03:54 ET | 37 pages

Equity Markets Positioning Model

Positioning Momentum Fades as Bond Volatility Pressures Risk Appetite

CITI'S TAKE

Global positioning remains constructive, but bullish flows are fading. Positioning activity across US indices has swung from accumulation towards long liquidation and new shorts over the past week, reversing previous positioning momentum support. Europe continues to exhibit a stronger positioning backdrop globally, although short activity is beginning to increase. In Asia, improving KOSPI positioning reflects a mix of flows including short-covering activity rather than outright bullish conviction. Taken together, the dominant theme suggests a transition towards a more selective and fragile risk environment. The key risks are profit-taking in long books, particularly Russell 2000, and potential squeeze-driven volatility in relatively larger offside short positions across Nikkei and KOSPI.

US – Bullish positioning weakens as bearish flows return — US positioning weakened as investor activity shifted towards long liquidation and new short positioning. Nasdaq experienced the largest regional weakening, with both S&P 500 and Nasdaq positioning drifting back towards neutral levels. Although aggregate positioning remains constructive, declining gross exposure suggests reduced investor participation. Russell 2000 remains the notable exception; with long exposure still near three-year highs, profit-taking risks remain elevated should bearish momentum persist.

Europe – Bullish positioning holds despite renewed short build — European positioning remains the strongest on aggregate among developed markets despite a modest deterioration in sentiment. Recent flows were driven primarily by new short positioning in EuroStoxx and DAX, while European Banks and FTSE witnessed increased long unwinds. Notably, positioning remains moderately bullish and well above neutral levels. The key tactical risk sits within DAX, where profitable longs coexist with deeply loss-making shorts, creating asymmetric flow dynamics. This setup leaves an ongoing risk of heightened volatility through either short squeezes or accelerated profit-taking activity.

Asia – Short covering and selective new long activity support KOSPI and Hang Seng — Positioning in Asia remains highly dispersed, but balance has improved over the past week. KOSPI and Hang Seng benefited from a combination of new risk flows and meaningful short covering, while several other regional markets continued to experience rising bearish flows. The most notable development is the growing vulnerability within the short books in both KOSPI and Nikkei, where rising losses increase the likelihood of squeeze risk becoming a more important driver of near-term index changes.

Europe

David T Chew^{AC}

+44-20-7986-7698

david.chew@citi.com

US

Richard W Schlatter

+1-212-816-0591

richard.w.schlatter@citi.com

Anju Bhandari

+1-212-816-3812

anju.bhandari@citi.com

Asia-Pacific

Yue Hin Pong

+44-20-7986-3953

yue.hin.pong@citi.com

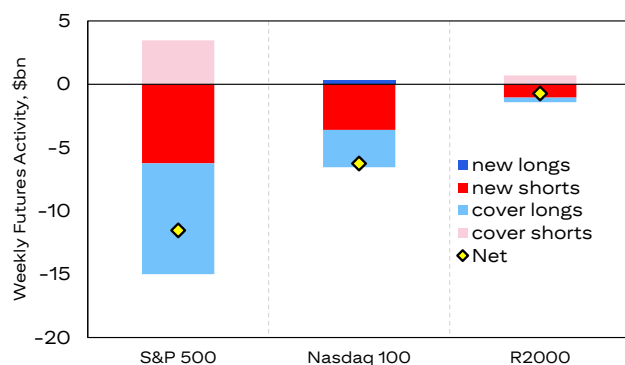
See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations.

Current Themes

US equities moved lower over the week as the sharp rise in long-end Treasury yields continued to weigh on risk assets. While the market found temporary support following the [US Treasury's surprise announcement](#) to expand long-term debt buybacks, rates volatility remained a key headwind for equities. At the same time, geopolitical uncertainty surrounding the potential expiry of the US-Iran ceasefire continues to warrant close monitoring.

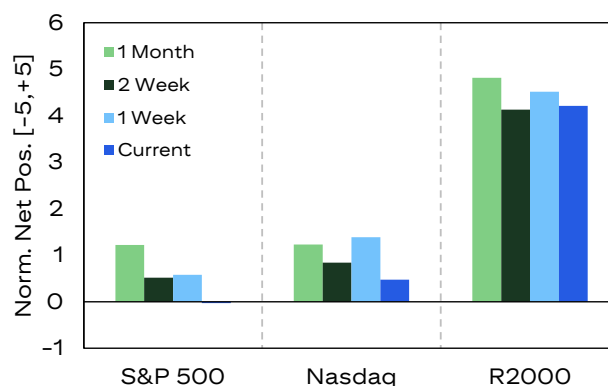
Against this backdrop, positioning dynamics have weakened. Higher yields appear consistent with the recent shift towards long liquidation and fresh short positioning across Nasdaq and the S&P, highlighting a reduced appetite for duration-sensitive growth exposures. This follows an already weakening positioning backdrop observed in the prior week. Normalised positioning for both the S&P 500 (0.0) and Nasdaq (+0.5) continue to drift back towards neutral levels, while the Russell 2000 (+4.2) remains the most extended market despite a modest easing in bullish exposure.

Figure 1. Bearish flow activity reverses momentum in US positioning rebuild.



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

Figure 2. US Large Cap positioning weakens as levels trend towards neutral.

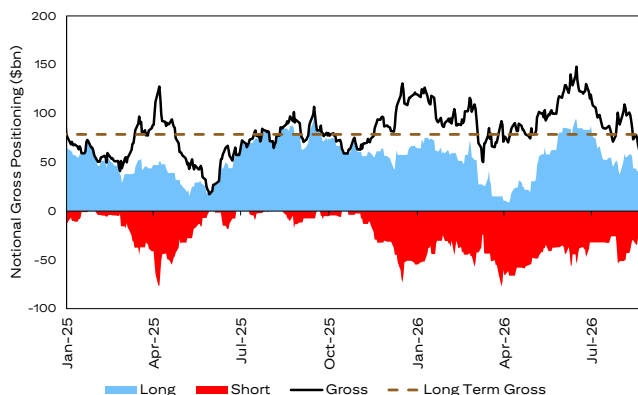


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

From a positioning risk perspective, profit/loss levels are relatively small. Equally, large cap gross positioning has been in decline (Figure 3, Figure 4), now falling back below long-term averages, suggesting weaker participation. The exception remains Russell 2000, with the long book notional still near 3-year highs. Positioning risk remains abnormally high, and this leaves the index more susceptible to profit-taking unwinds if bearish momentum continues. A decline below 2,900 would move the majority of legacy long positions into unrealised

losses, increasing the risk of additional position reduction and reinforcing downside pressure.

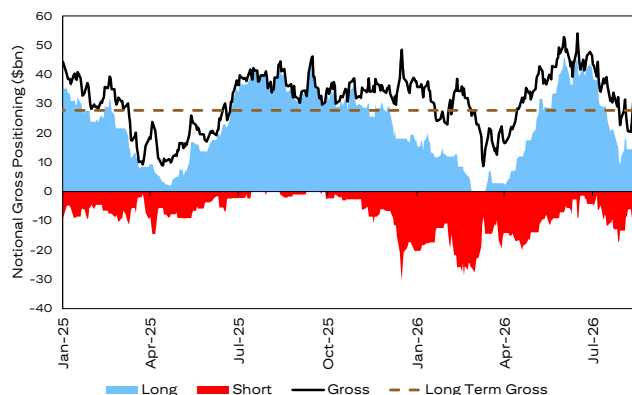
Figure 3. S&P 500 gross exposure contracts as positioning momentum fades



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

Figure 4. Nasdaq gross positioning follows a similar path



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

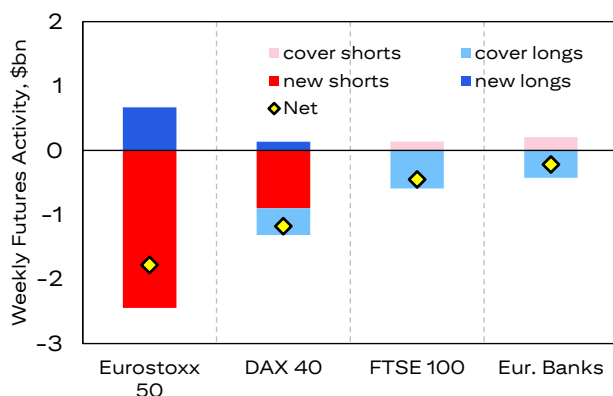
Source: Citi Research

Europe

European markets followed recent US with almost all indices in
the region declining over the week, as continued geopolitical tension, alongside bond market volatility, added a headwind to further outperformance. FTSE 100 stood out as an outlier, with the index outperforming peers, benefiting from the uplift in oil commodity prices—the same dynamic that was seen hurting continental European markets.

EuroStoxx (+1.9) and DAX (+1.3) flows were predominantly led by new short positioning, whereas FTSE (+1.6) and Euro Banks (+1.4) saw increased unwinds of long positioning. Bullish positioning was easing across all European indices albeit at the margin, and despite the decline seen in recent days, net positioning remains moderately bullish (Figure 6).

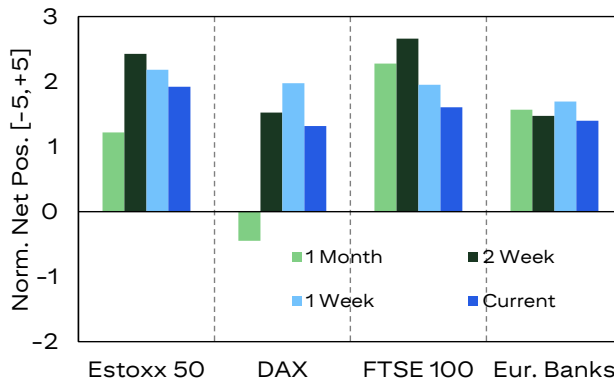
Figure 5. Weekly activity turns bearish for Europe...



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

Figure 6. ...but positioning remains resilient despite flow deterioration



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

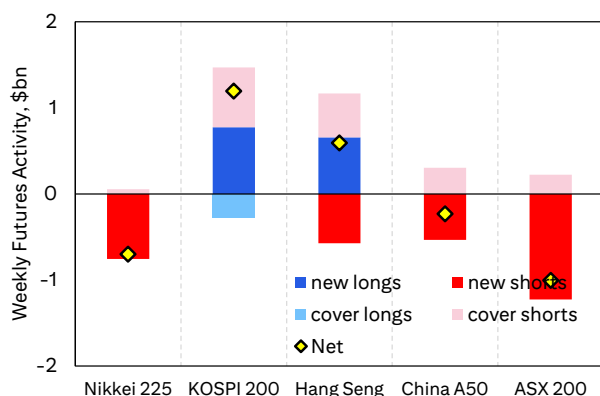
The DAX presents a particularly interesting P&L story. The composition of deeply loss-making shorts alongside a profitable long book is pronounced, and this polarization may increase the potential for elevated near-term

volatility, as either short-covering demand or profit-taking activity could become the dominant flow driver.

Asia/EM

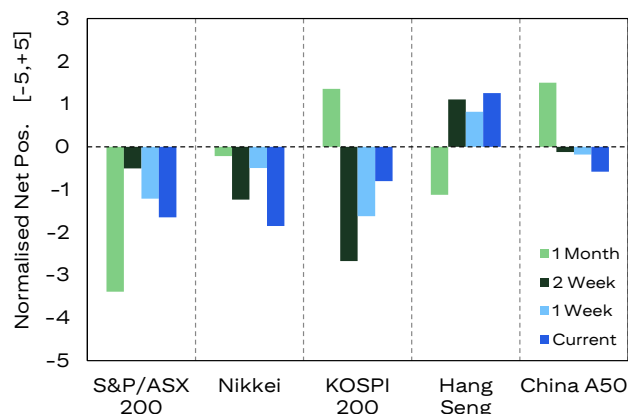
Positioning in Asian markets continues to reflect one of considerable dispersion, with KOSPI (-0.8) and Hang Seng (+1.3) positioning improving through a combination of modest new long activity and meaningful short covering. Other indices in the region continue to see a decline from new short positioning.

Figure 7. KOSPI and Hang Seng continue to recover as trends diverge across Asia



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

Figure 8. Positioning continues to improve for KOSPI and Hang Seng



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

Rising losses in the Nikkei and KOSPI short books leave scope for additional short-covering demand should market momentum remain supportive. The offside risk for shorts in both markets is high, suggesting that any positive news could trigger a significant squeeze.

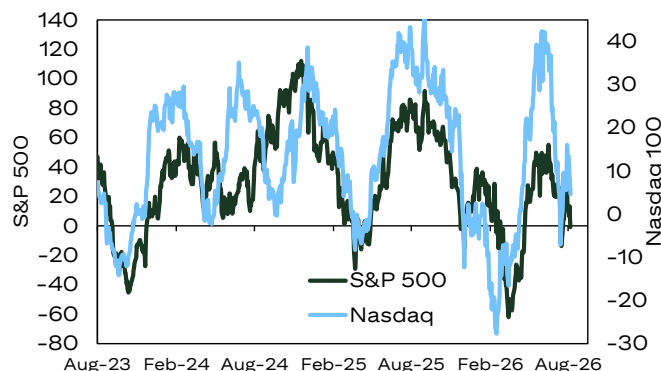
Figure 9. Positioning change across global equities

	By 1w change		
	Change	Pos	PnL
KOSPI 200	0.8	-0.8	0.0
MSCI EM	0.6	2.1	-0.4
Hang Seng	0.4	1.3	1.5
EuroStoxx	-0.3	1.9	0.3
Eur Banks	-0.3	1.4	0.4
R2000	-0.3	4.2	1.2
FTSE	-0.3	1.6	0.4
China A50	-0.4	-0.6	0.0
ASX 200	-0.4	-1.6	0.2
S&P Mini	-0.6	0.0	-0.4
DAX	-0.7	1.3	-1.1
Nasdaq	-0.9	0.5	-0.4
Nikkei	-1.4	-1.8	-0.2
DM xUS	-1.8	0.5	1.0

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

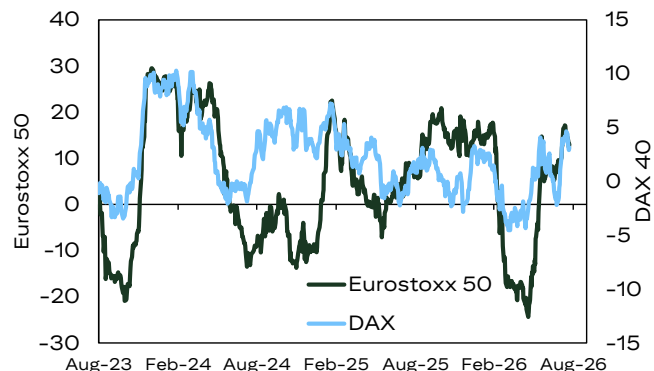
How Has Positioning Changed?

Figure 10. S&P 500 (ES) and Nasdaq 100 (NQ) net positioning



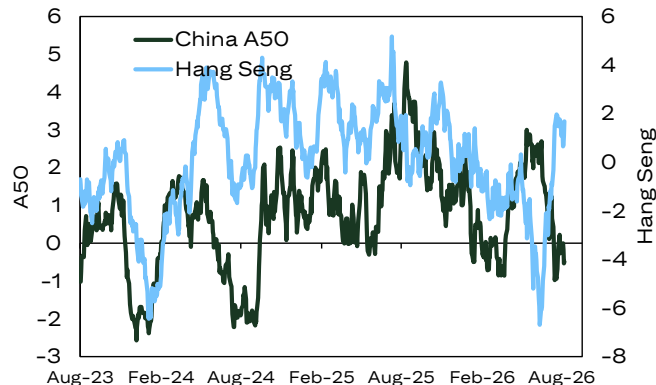
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Bloomberg

Figure 11. EuroStoxx 50 (VG) and DAX 40 (GX) net positioning



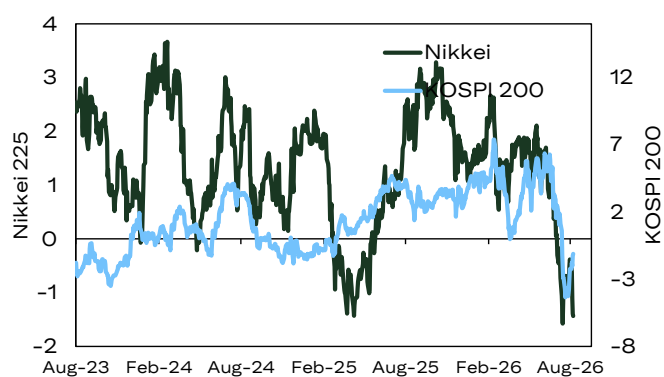
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Bloomberg

Figure 12. Hang Seng (HI) and China A50 (XU) net positioning



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Bloomberg

Figure 13. Nikkei 225 (NI) and KOSPI 200 (KM) net positioning



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Bloomberg

Figure 14. Ranking of equity market positioning

21/08/2026	By Position	PnL	By 1w change	Change	Pos	PnL	By PnL	PnL	Pos	By Position	Percentile
R2000	4.2	1.2	KOSPI 200	0.8	-0.8	0.0	Hang Seng	1.5	1.3	R2000	98%
MSCI EM	2.1	-0.4	MSCI EM	0.6	2.1	-0.4	R2000	1.2	4.2	FTSE	79%
EuroStoxx	1.9	0.3	Hang Seng	0.4	1.3	1.5	DM xUS	1.0	0.5	MSCI EM	75%
FTSE	1.6	0.4	EuroStoxx	-0.3	1.9	0.3	FTSE	0.4	1.6	Hang Seng	73%
Eur Banks	1.4	0.4	Eur Banks	-0.3	1.4	0.4	Eur Banks	0.4	1.4	EuroStoxx	66%
DAX	1.3	-1.1	R2000	-0.3	4.2	1.2	EuroStoxx	0.3	1.9	Eur Banks	66%
Hang Seng	1.3	1.5	FTSE	-0.3	1.6	0.4	ASX 200	0.2	-1.6	DAX	64%
DM xUS	0.5	1.0	China A50	-0.4	-0.6	0.0	KOSPI 200	0.0	-0.8	DM xUS	52%
Nasdaq	0.5	-0.4	ASX 200	-0.4	-1.6	0.2	China A50	0.0	-0.6	Nasdaq	33%
S&P Mini	0.0	-0.4	S&P Mini	-0.6	0.0	-0.4	Nikkei	-0.2	-1.8	China A50	17%
China A50	-0.6	0.0	DAX	-0.7	1.3	-1.1	Nasdaq	-0.4	0.5	S&P Mini	17%
KOSPI 200	-0.8	0.0	Nasdaq	-0.9	0.5	-0.4	MSCI EM	-0.4	2.1	KOSPI 200	17%
ASX 200	-1.6	0.2	Nikkei	-1.4	-1.8	-0.2	S&P Mini	-0.4	0.0	ASX 200	16%
Nikkei	-1.8	-0.2	DM xUS	-1.8	0.5	1.0	DAX	-1.1	1.3	Nikkei	0%

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Bloomberg

Figure 15. Summary

		S&P 500	Nasdaq	R2000	Estoxx 50	DAX	FTSE 100	Eur. Banks	S&P/ASX 200	Nikkei	MSCI EAFE	MSCI EM	KOSPI 200	Hang Seng	China A50
(\$bn)	21/08/2026	ES	NQ	RTY	VG	GX	Z	CA	XP	NI	MFS	MES	KM	HI	XU
Equity	Net Notional	-0.7	4.6	10.3	13.1	3.1	4.7	1.8	-3.1	-1.4	0.8	4.9	-1.1	1.7	-0.5
	Percentile	17%	33%	98%	66%	64%	79%	66%	16%	0%	52%	75%	16.6%	73%	17%
Shorts	Notional	-29.3	-10.3	-2.4	-7.3	-8.3	-2.2	0.0	-4.0	-2.3	-2.7	-2.1	-2.1	-1.1	-0.5
	Percentile	58%	75%	35%	54%	98%	70%	5%	75%	95%	54%	43%	80%	45%	45%
	Avg. Entry	7,594	29,500	3,020	6,373	25,171	10,796	--	8,991	66,038	3,251.1	1,615	1,074.0	25,406	14,768
	% offside	70%	28%	49%	73%	90%	18%	--	56%	53%	40%	100%	61%	100%	65%
Longs	Notional	28.5	14.9	12.7	20.4	11.4	6.9	1.8	0.9	0.9	3.5	6.9	1.0	2.8	0.0
	Percentile	16%	34%	99%	77%	99%	85%	46%	22%	4%	51%	71%	15%	72%	4%
	Avg. Entry	7,686	29,751	2,995	6,404	25,599	10,743	312.0	8,902	68,008	3,197	1,704	1,055.1	24,549.8	--
	% offside	53%	91%	26%	42%	24%	40%	39%	37%	92%	21%	31%	0%	21%	--
Activity	Ref Price	7,691	29,375	3,017	6,465	26,133	10,817	317.2	8,999	65,732	3,254	1,719	1,085.5	25,837.9	14,786
	1-day Type	6.0	-2.9	0.6	0.7	-0.1	-0.1	0.2	-0.3	-0.2	0.0	0.0	0.2	0.6	0.0
	1-week	↓ -11.5	↓ -6.2	↓ -0.7	↓ -1.8	↓ -1.2	↓ -0.5	↓ -0.2	↓ -1.0	↓ -0.7	↓ -2.9	↓ 1.8	↓ 1.2	↓ 0.6	↓ -0.2
Normalised Scores (-5 / +5)															
Positioning	Equity implied	-0.0	0.5	4.2	1.9	1.3	1.6	1.4	-1.7	-1.9	0.5	2.1	-0.8	1.3	-0.6
	ETF	2.1	2.8		-1.0	-1.1	-0.9	0.8		0.4		1.1			
	3m Net Pos	1.0	1.7	4.2	0.5	0.1	0.3	1.1	-1.7	-0.7	0.5	1.6	-0.8	1.3	-0.6
	1w Change	-0.8	-1.2	-0.3	0.0	-0.3	-0.2	0.1	-0.4	-1.3	-1.8	0.3	0.8	0.4	-0.4
P&L	Equity implied	-0.4	-0.4	1.2	0.3	-1.1	0.4	0.4	0.2	-0.2	1.0	-0.4	0.0	1.5	-0.0
	ETF	1.0	-1.4		-1.1	-0.4	-1.3	0.1		1.3		0.9			
	3m Net PnL	0.3	-0.9	1.2	-0.4	-0.8	-0.5	0.2	0.2	0.5	1.0	0.2	0.0	1.5	-0.0
	1w Change	-1.1	-1.1	-2.1	-0.1	-0.5	-0.1	0.0	0.7	-0.0	-0.5	0.3	0.2	0.4	-0.0

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

Figure 16. Average Entry Price Levels and Profit Breakeven Levels

		S&P 500	Nasdaq	R2000	Estoxx 50	DAX	FTSE 100	Eur. Banks	S&P/ASX 200	Nikkei	MSCI EAFE	MSCI EM	KOSPI 200	Hang Seng	China A50
	21/08/2026	ES	NQ	RTY	VG	GX	Z	CA	XP	NI	MFS	MES	KM	HI	XU
Notional	\$bn	-0.7	4.6	10.3	13.1	3.1	4.7	1.8	-3.1	-1.4	0.8	4.9	-1.1	1.7	-0.5
	(-5/+5)	-0.0	0.5	4.2	1.9	1.3	1.6	1.4	-1.7	-1.9	0.5	2.1	-0.8	1.3	-0.6
PnL	\$bn	-0.4	-0.1	0.1	0.1	-0.1	0.0	0.0	0.0	-0.0	0.1	-0.1	0.0	0.1	-0.0
	(-5/+5)	-0.4	-0.4	1.2	0.3	-1.1	0.4	0.4	0.2	-0.2	1.0	-0.4	0.0	1.5	-0.0
Ref Price		7,691	29,375	3,017	6,465	26,133	10,817	317.2	8,999	65,732	3,254	1,719	1,086	25,838	14,786
Shorts	Size (\$bn)	-29.3	-10.3	-2.4	-7.3	-8.3	-2.2	0.0	-4.0	-2.3	-2.7	-2.1	-2.1	-1.1	-0.5
	Avg Price	7,594	29,500	3,020	6,373	25,171	10,796	--	8,991	66,038	3,251	1,615	1,074.0	25,406	14,768
	Avg P&L	-1.3%	0.4%	0.1%	-1.4%	-3.7%	-0.2%	--	-0.1%	0.5%	-0.1%	-6.0%	-1.1%	-1.7%	-0.1%
	Offside (%)	70%	28%	49%	73%	90%	18%	--	56%	53%	40%	100%	61%	100%	65%
- Offside price															
	100%	7,788	29,702	3,063	6,568	26,472	10,858	--	9,191	68,901	3,268	1,671	1,184.0	25,555	14,835
	75%	7,666	29,627	3,041	6,447	25,149	10,853	--	9,052	67,389	3,263	1,625	1,083.9	25,442	14,761
	50%	7,542	29,391	3,011	6,299	25,070	10,841	--	8,993	65,423	3,257	1,592	1,057.7	25,311	14,733
	25%	7,512	29,375	2,975	6,272	24,966	10,798	--	8,866	64,774	3,200	1,579	1,025.1	25,224	14,733
	0%	7,454	29,374	2,975	6,258	24,873	10,509	--	8,831	64,684	3,137	1,577	983.8	25,173	14,733
Longs	Size (\$bn)	28.5	14.9	12.7	20.4	11.4	6.9	1.8	0.9	0.9	3.5	6.9	1.0	2.8	0.0
	Avg Price	7,686	29,751	2,995	6,404	25,599	10,743	312.0	8,902	68,008	3,197	1,704	1,055.1	24,550	--
	Avg P&L	0.1%	-1.3%	0.7%	0.9%	2.1%	0.7%	1.7%	1.1%	-3.3%	1.8%	0.8%	2.9%	5.2%	--
	Offside (%)	53%	91%	26%	42%	24%	40%	39%	37%	92%	21%	31%	0%	21%	--
- Offside price															
	100%	7,600	28,697	2,908	6,258	24,792	10,533	300.0	8,756	64,933	3,164	1,649	1,023.6	23,125	--
	75%	7,600	29,205	2,965	6,293	25,160	10,631	302.5	8,778	65,414	3,164	1,687	1,023.6	23,440	--
	50%	7,620	29,686	2,994	6,345	25,545	10,702	308.5	8,778	68,674	3,170	1,699	1,037.3	23,570	--
	25%	7,780	29,917	3,017	6,505	25,895	10,840	323.5	8,911	69,616	3,209	1,724	1,051.9	25,611	--
	0%	7,782	30,116	3,067	6,577	26,533	10,915	326.7	9,192	70,208	3,271	1,747	1,066.6	25,838	--

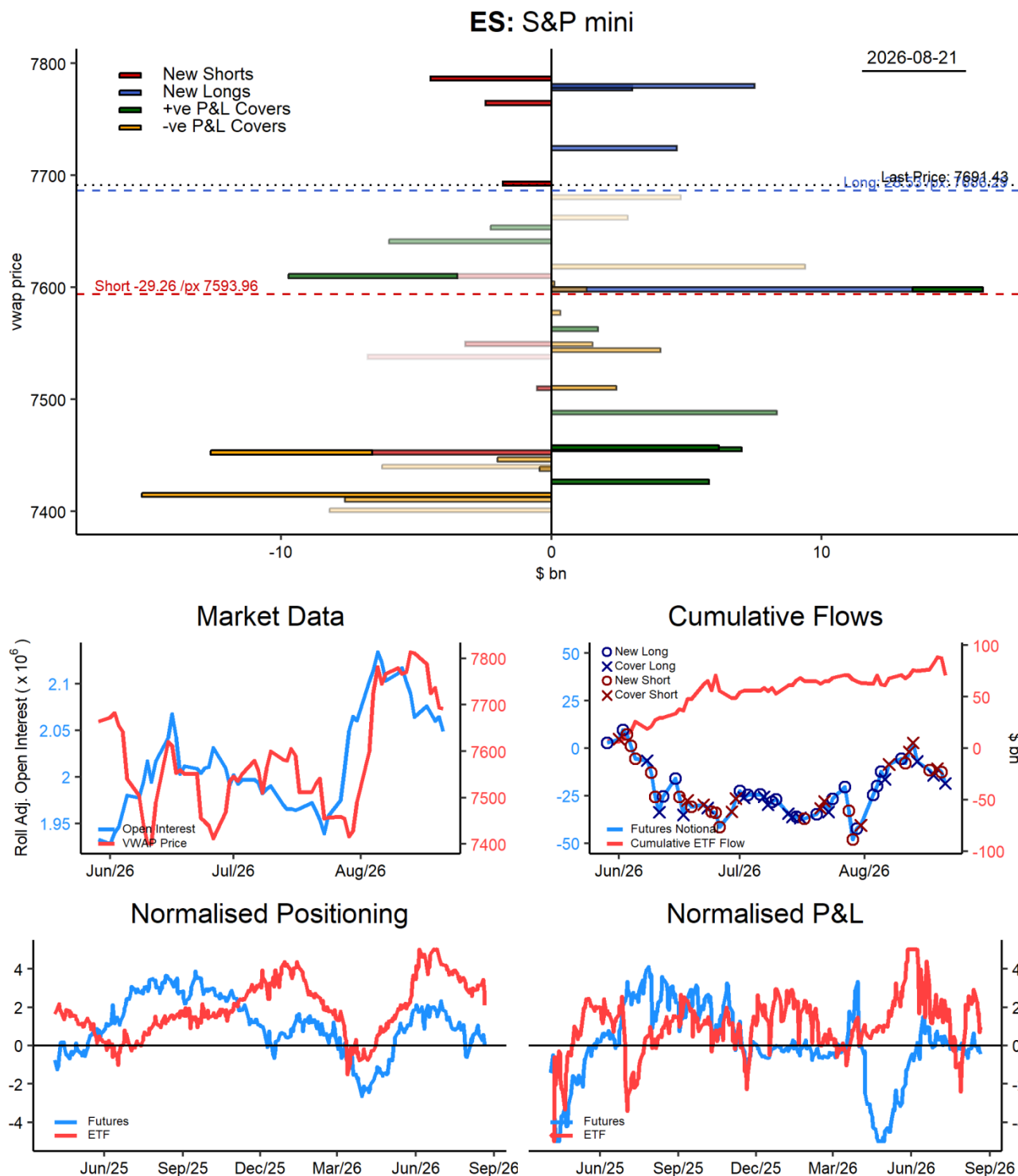
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

Equity Markets Positioning Charts

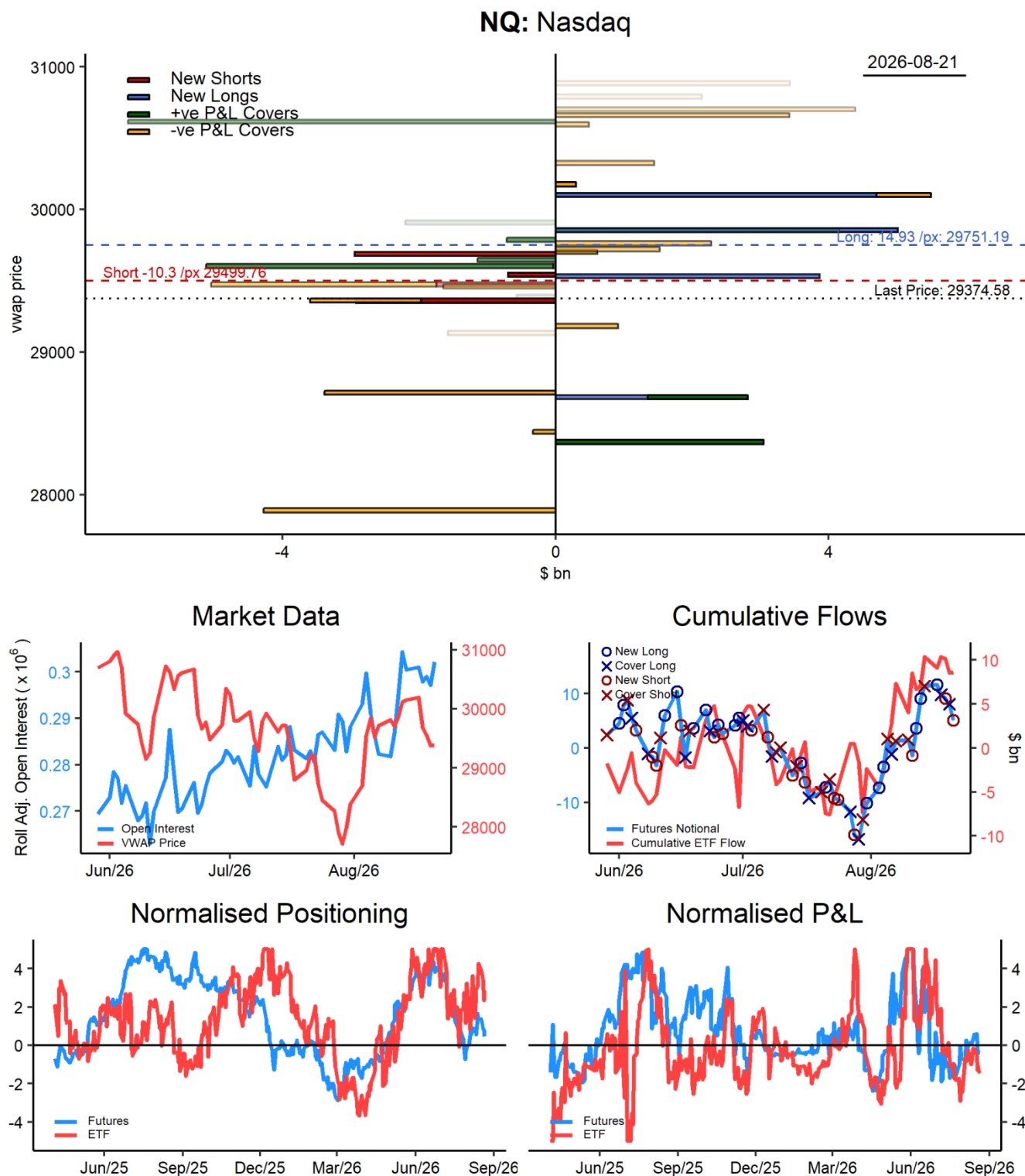
S&P 500

Figure 17. S&P 500 mini contract (ES) – position snapshot and time-series



NASDAQ 100

Figure 18. Nasdaq 100 contracts (NQ) – position snapshot and time-series

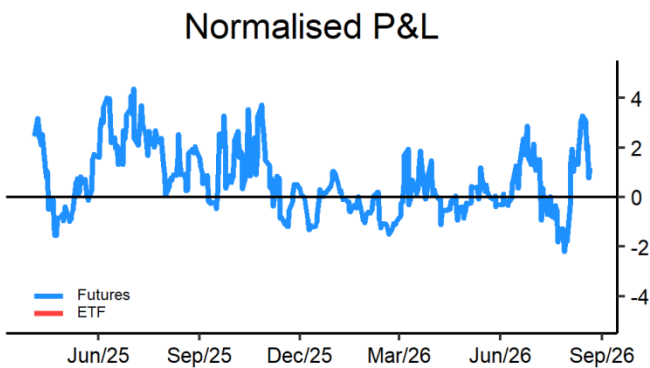
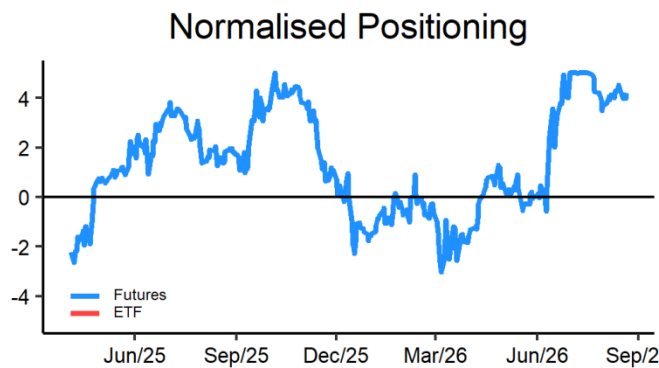
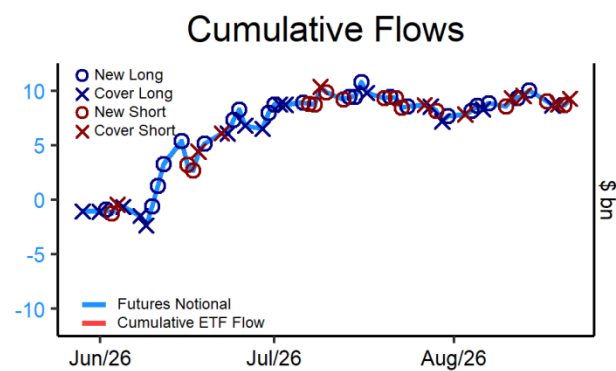
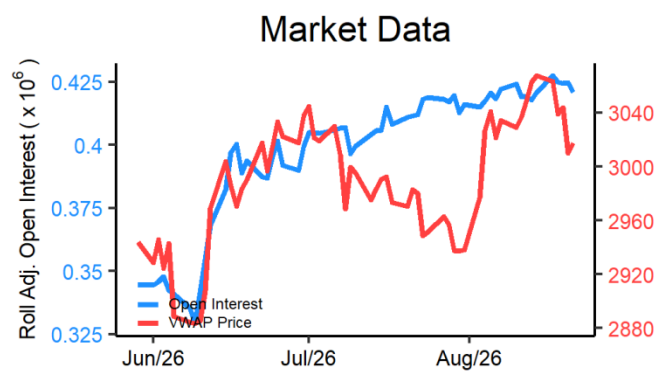
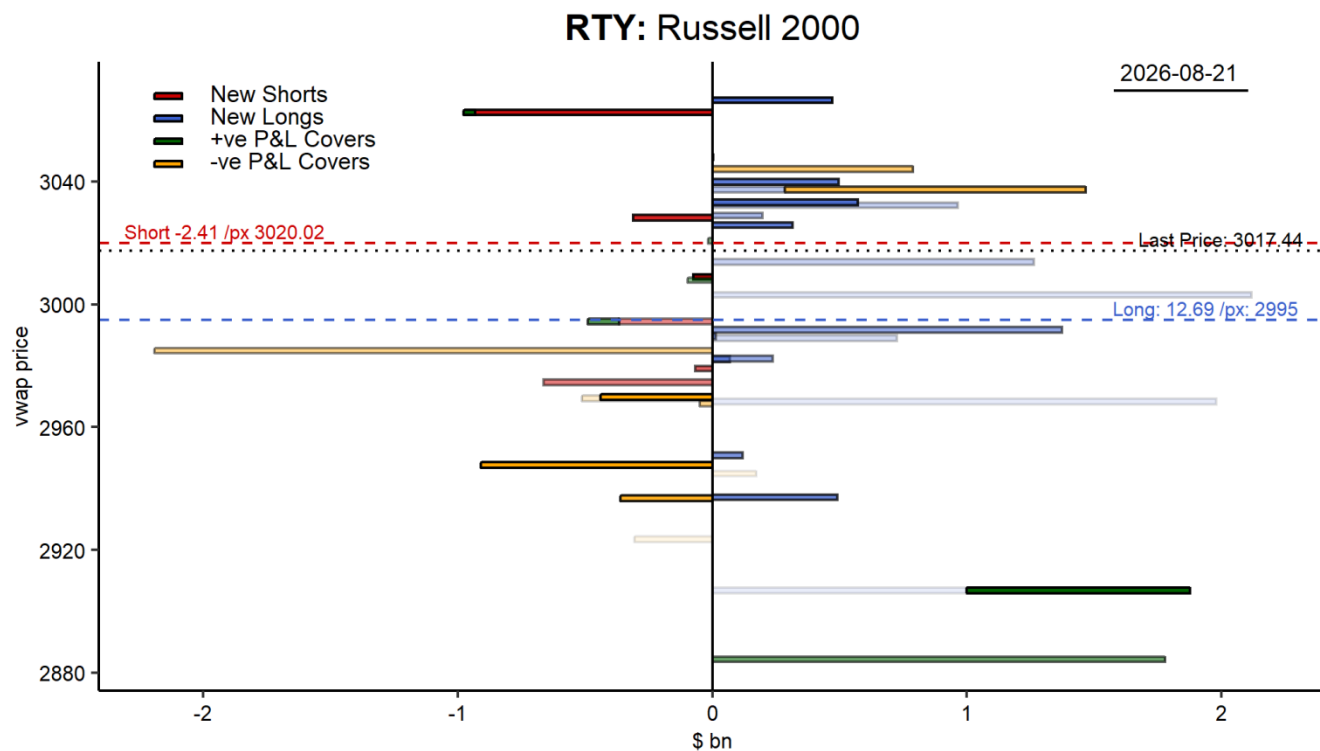


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

Russell 2000

Figure 19. Russell 2000 contracts (RTY) – position snapshot and time-series

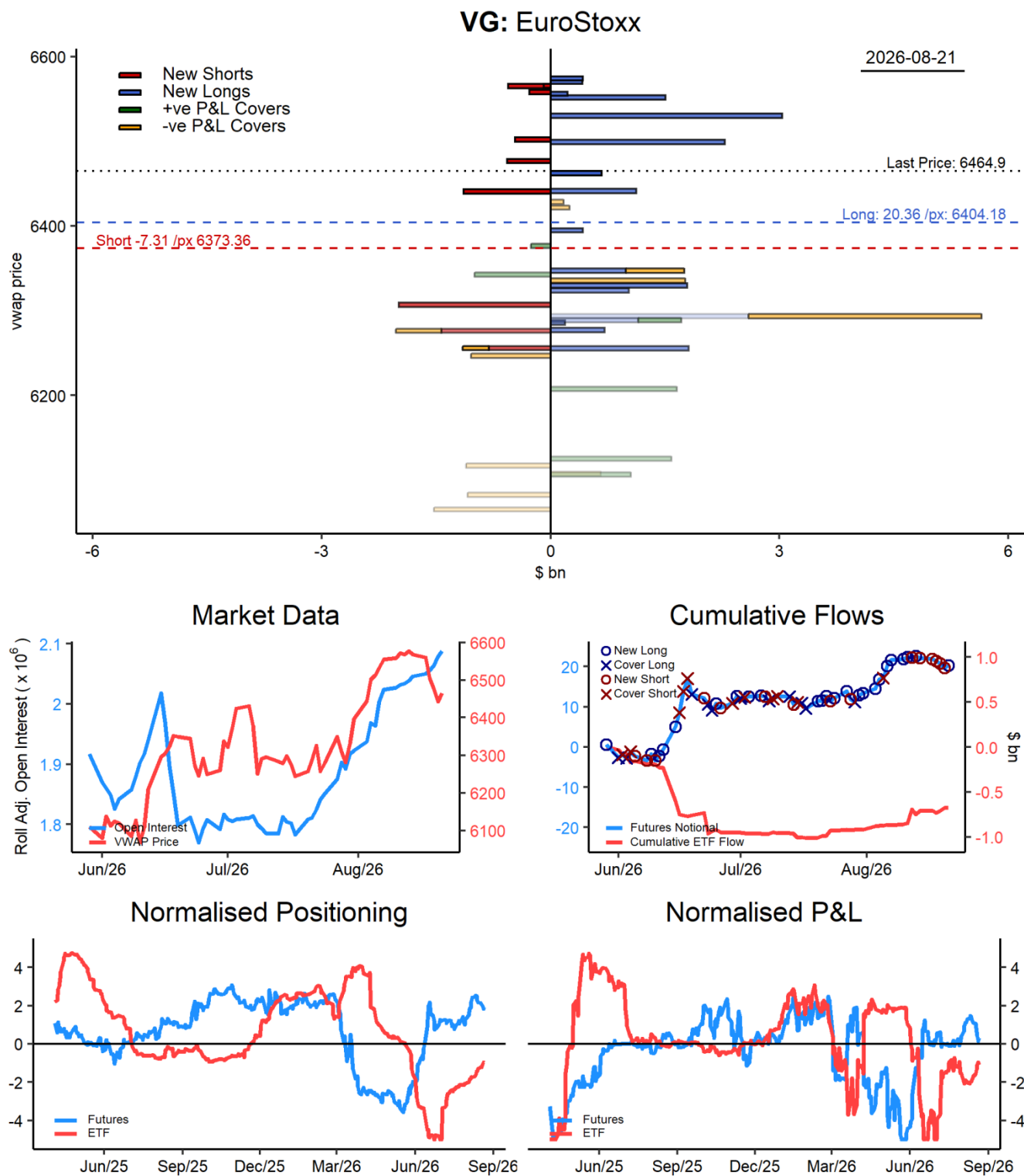


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

EuroStoxx 50

Figure 20. Eurostoxx 50 contracts (VG) – position snapshot and time-series

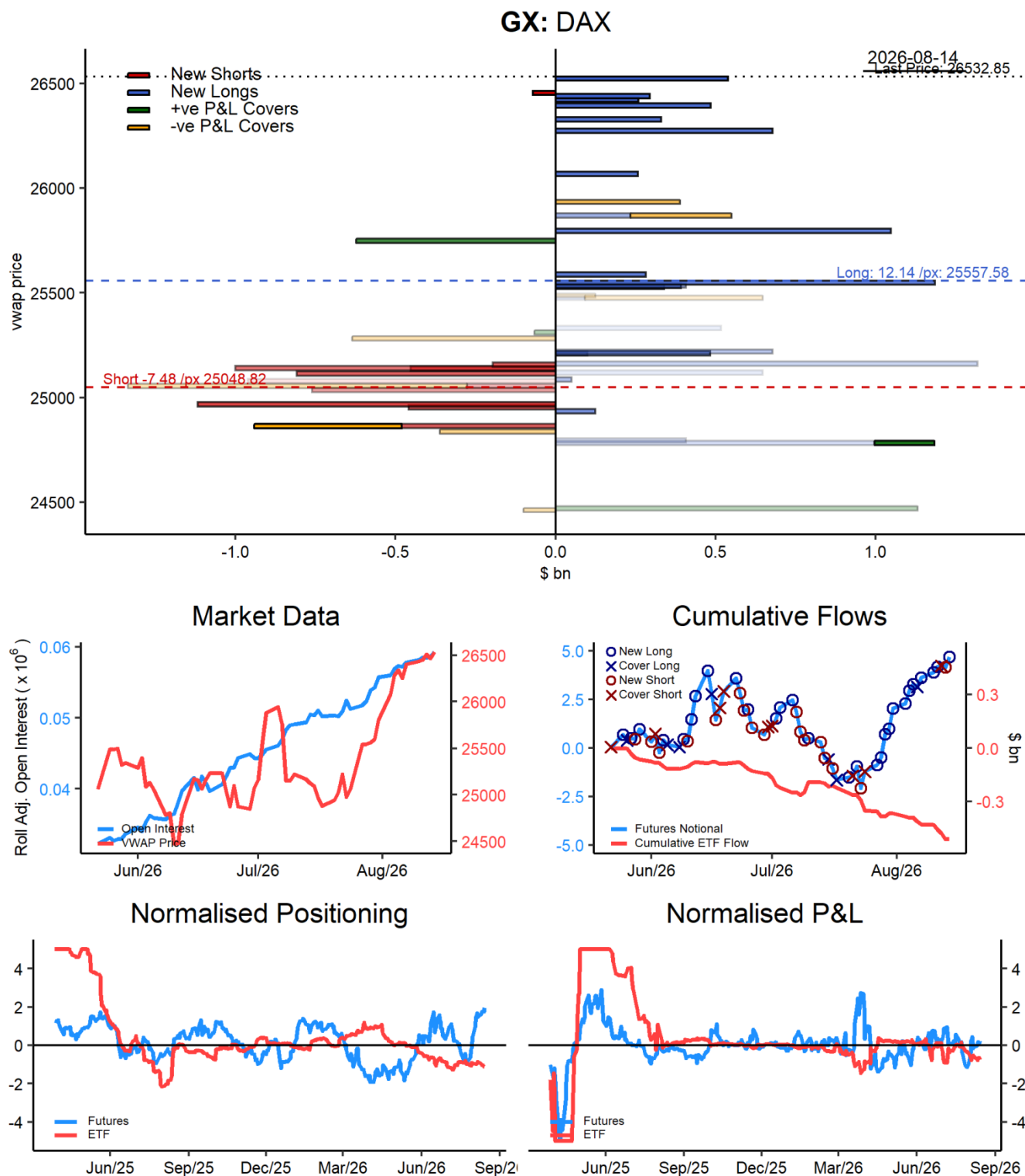


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

DAX

Figure 21. DAX contracts (GX) – position snapshot and time-series

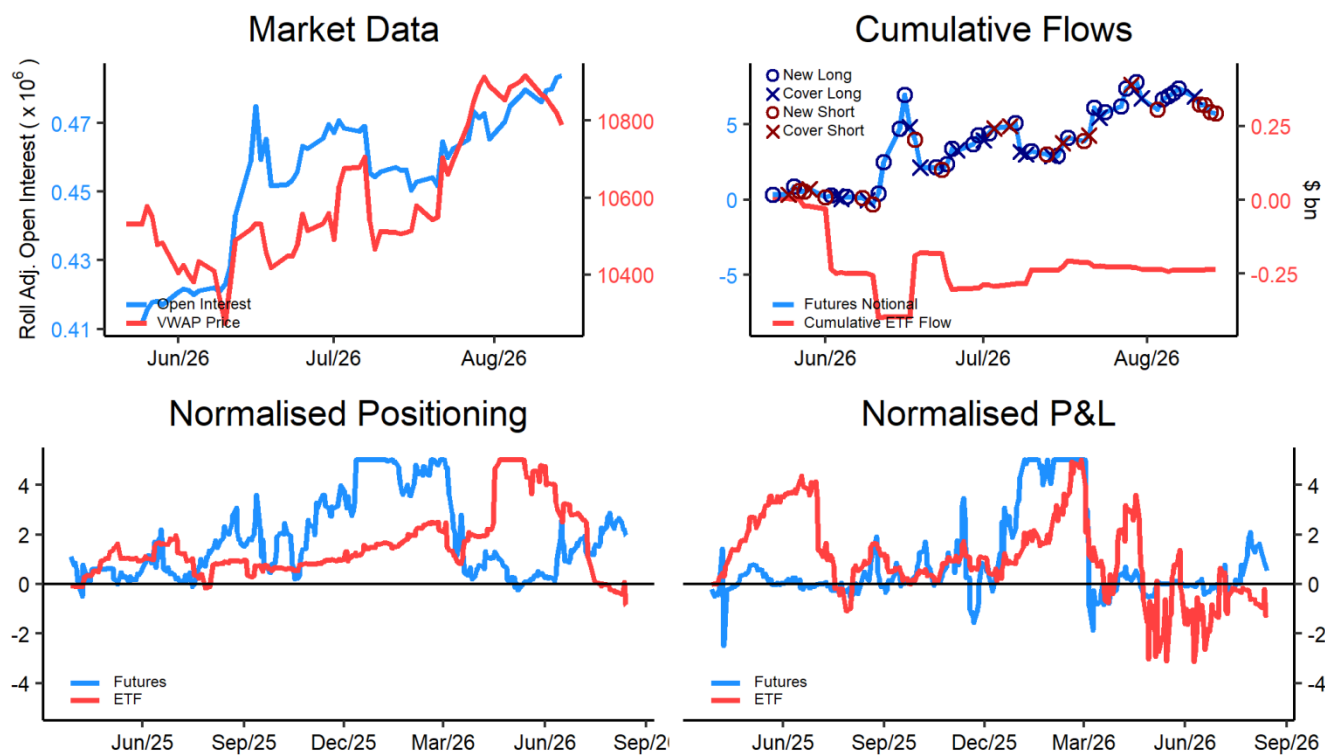
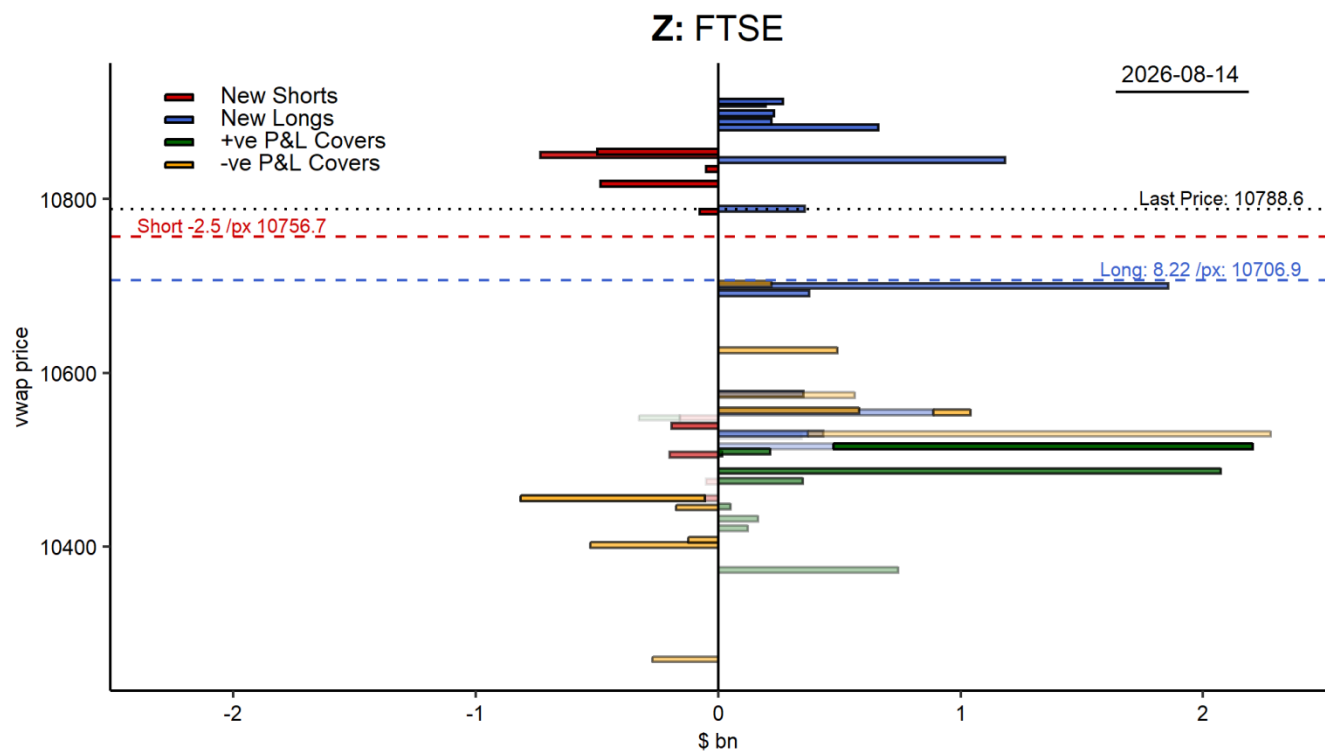


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

FTSE 100

Figure 22. FTSE 100 contracts (Z) – position snapshot and time-series

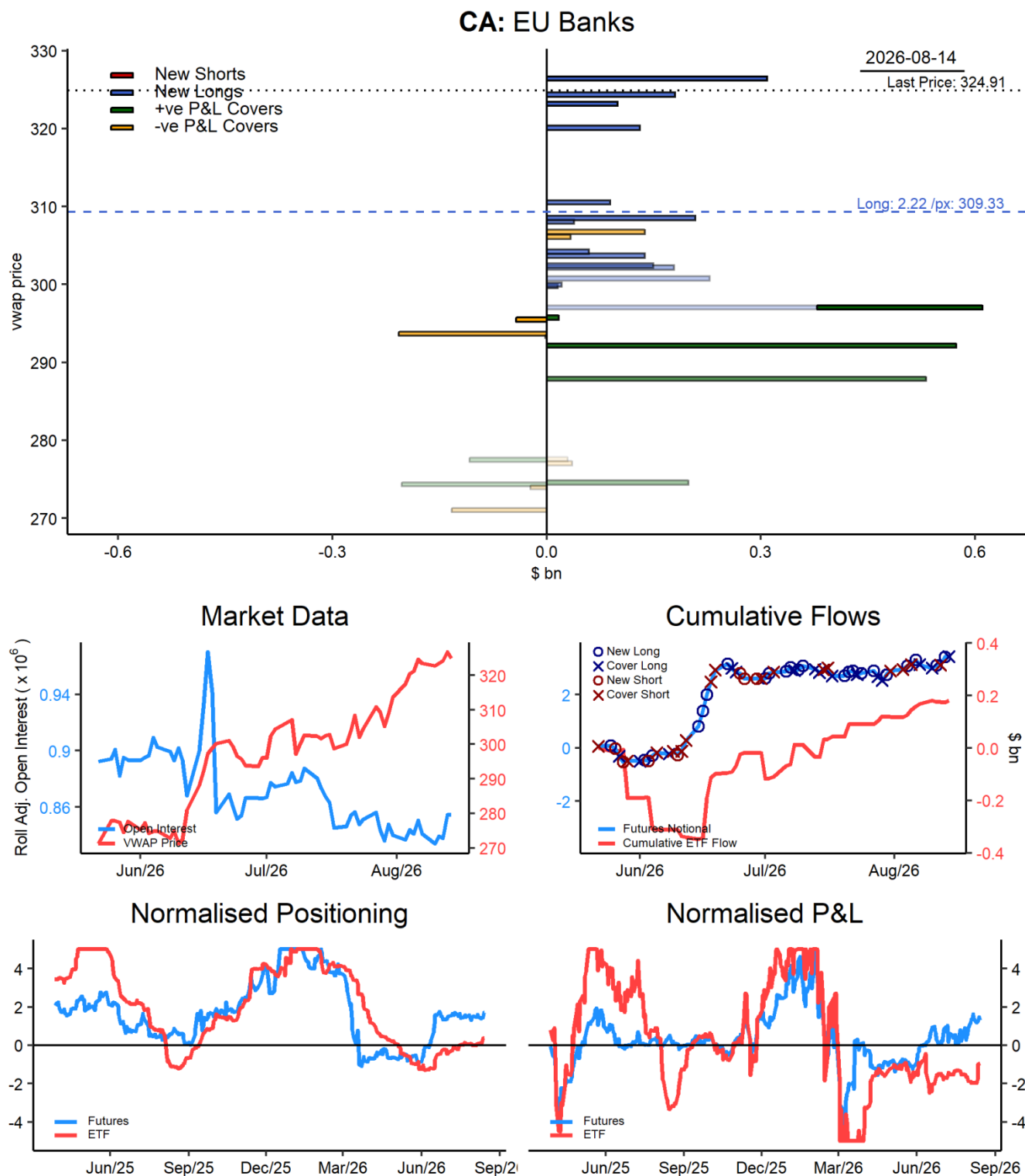


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

European Banks (SX7E)

Figure 23. Eurostoxx Banks (SX7E) Equity contracts (CA) – position snapshot and time-series

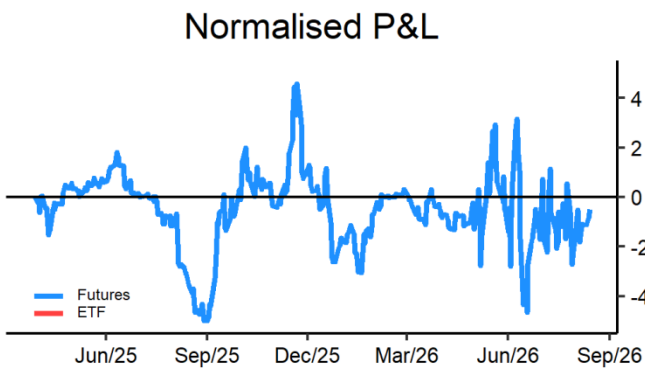
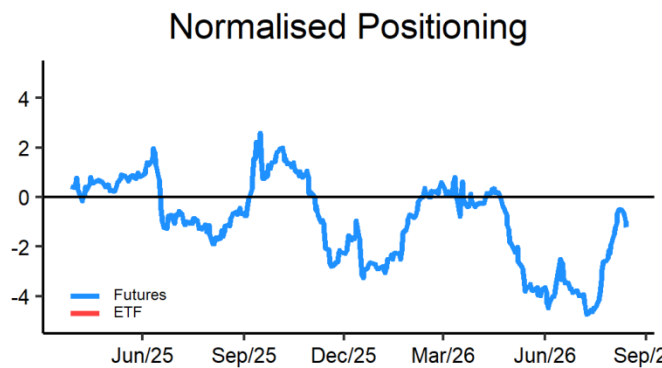
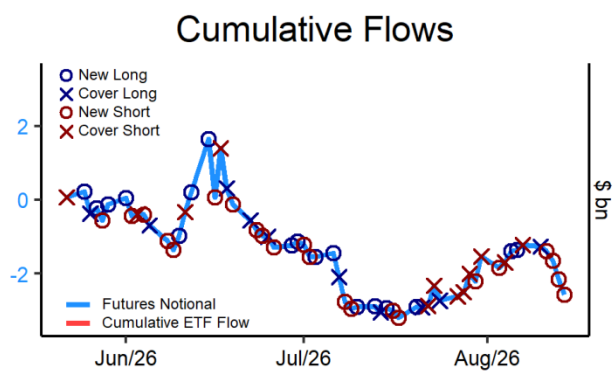
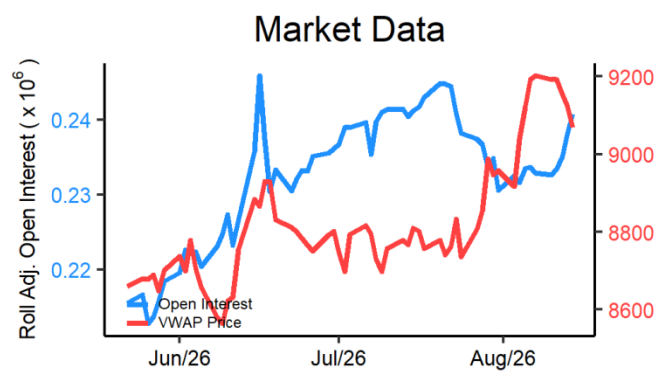
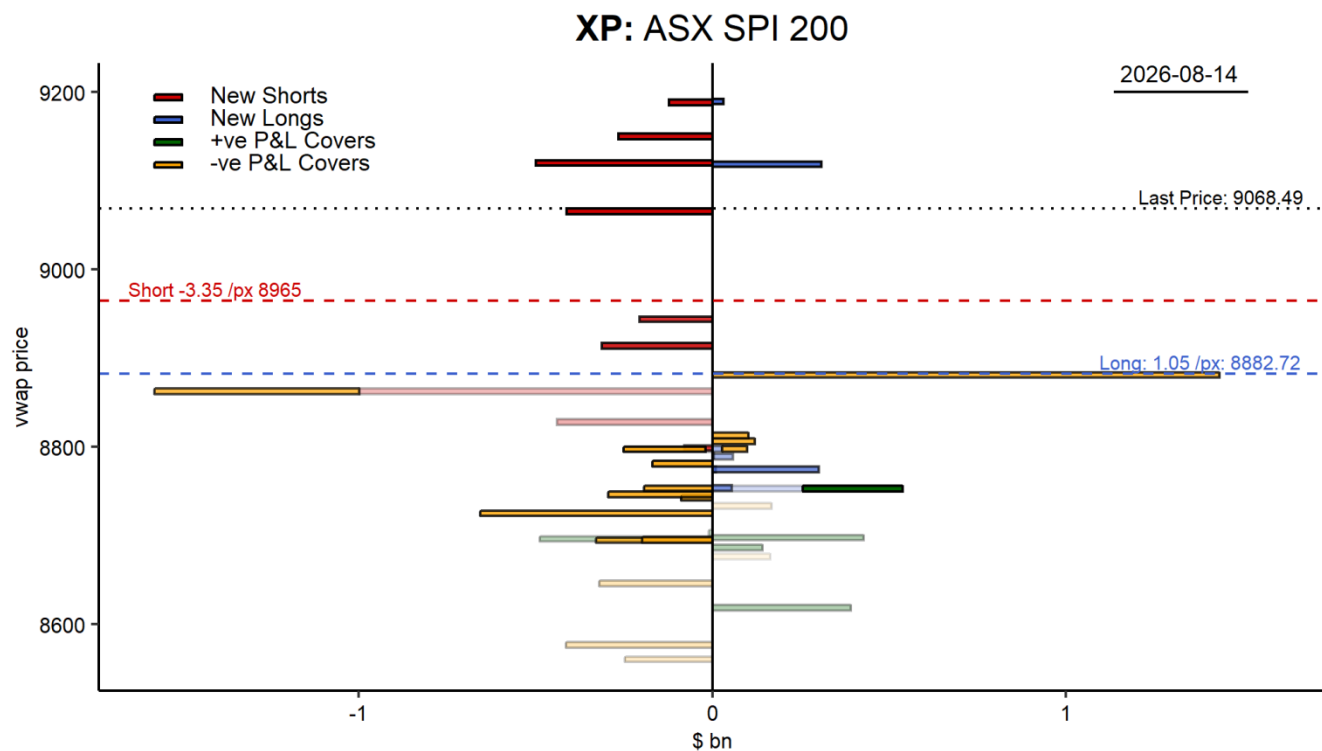


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

S&P ASX 200

Figure 24. S&P ASX 200 Equity contracts (XP) – position snapshot and time-series

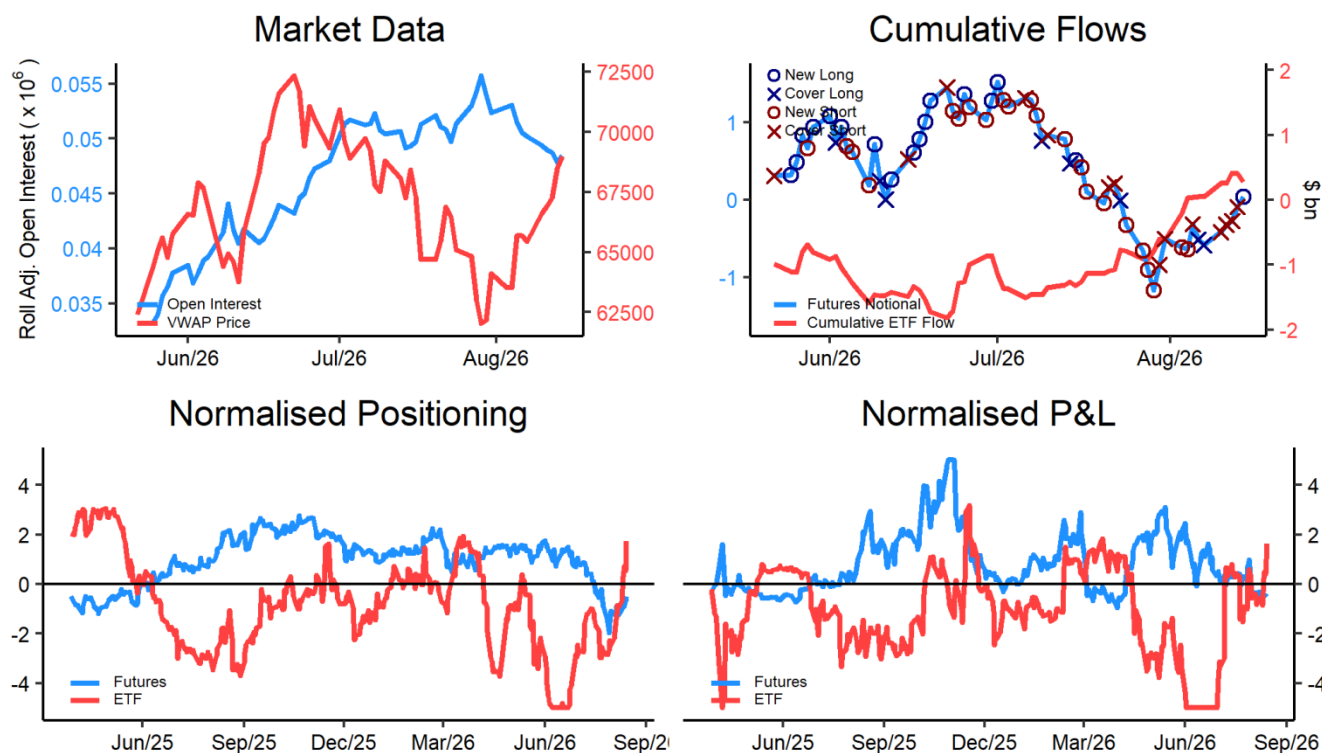
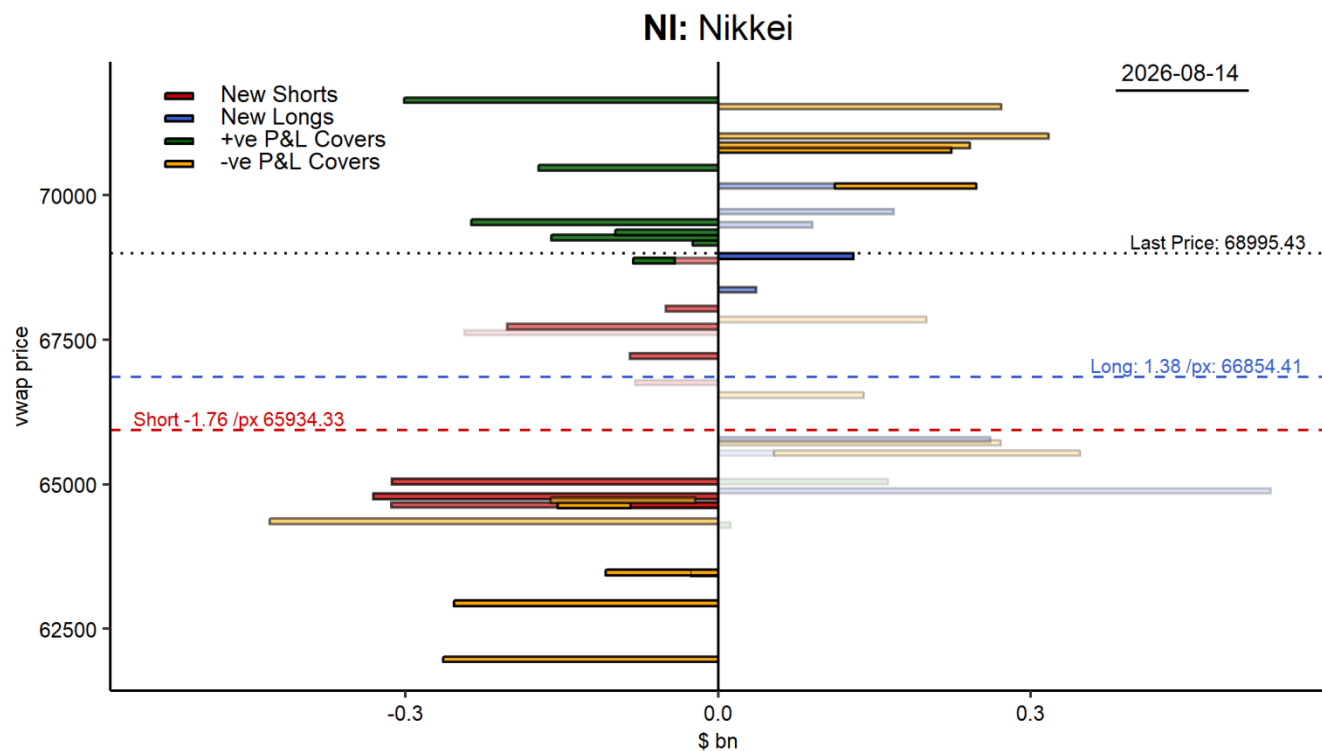


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

Nikkei 225

Figure 25. Nikkei contracts (NI) – position snapshot and time-series

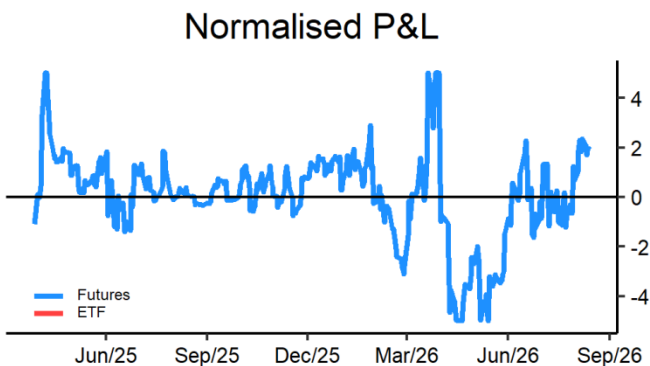
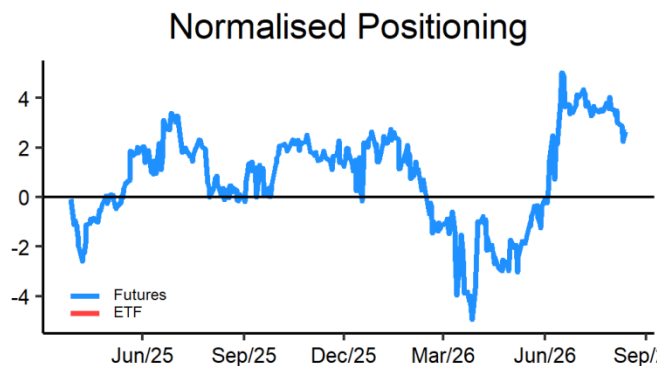
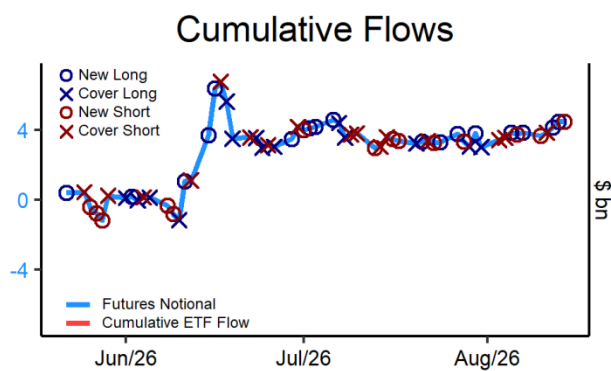
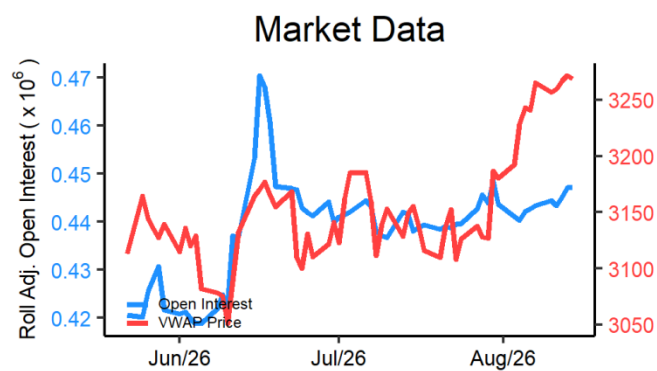
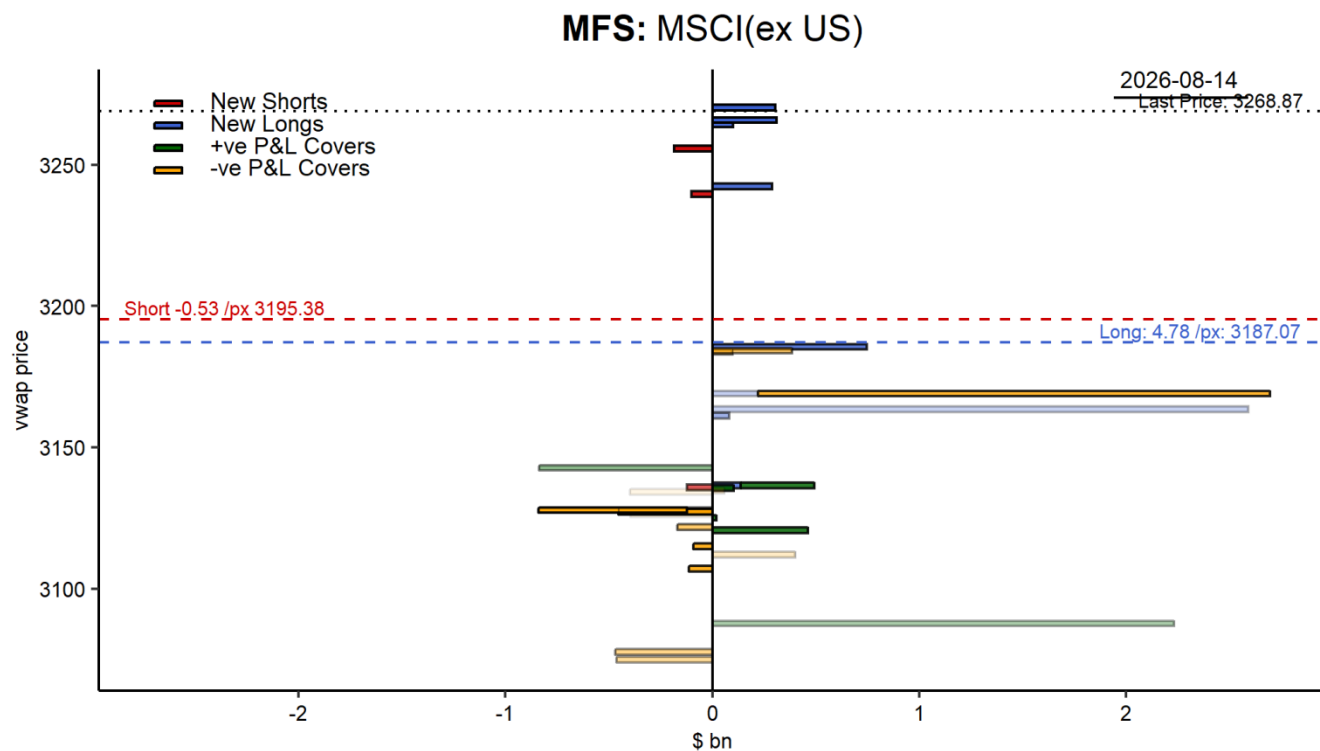


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

MSCI EAFE

Figure 26. MSCI EAFE (DM ex-US) (MFS) – position snapshot and time-series

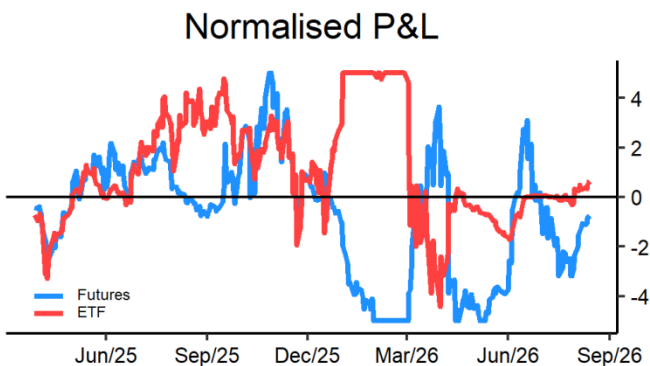
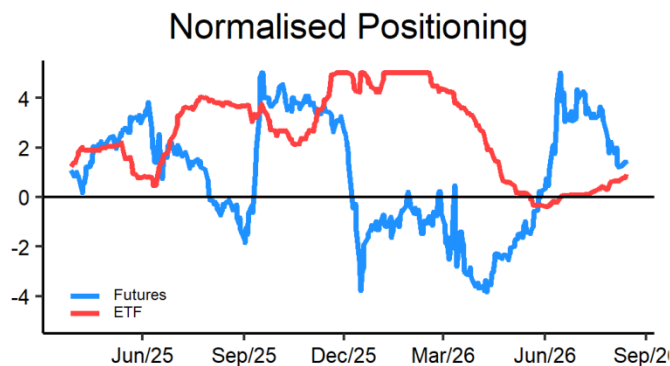
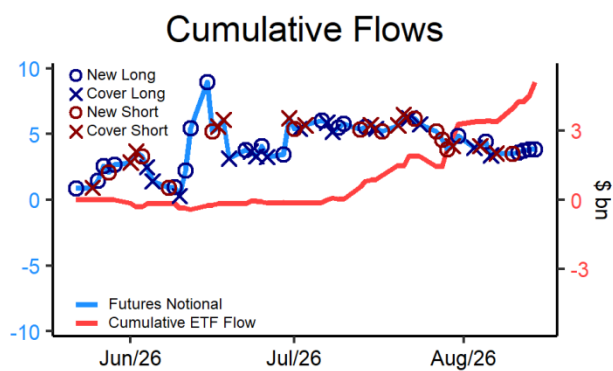
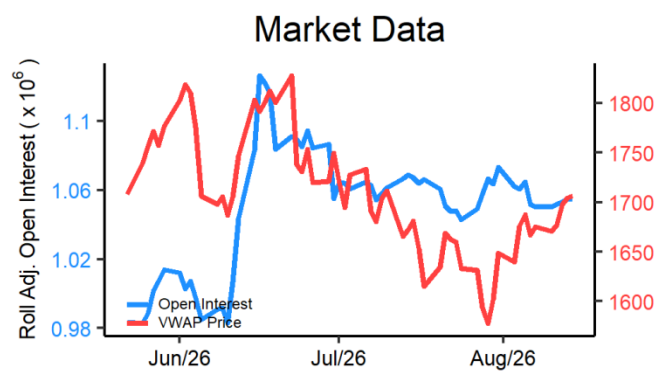
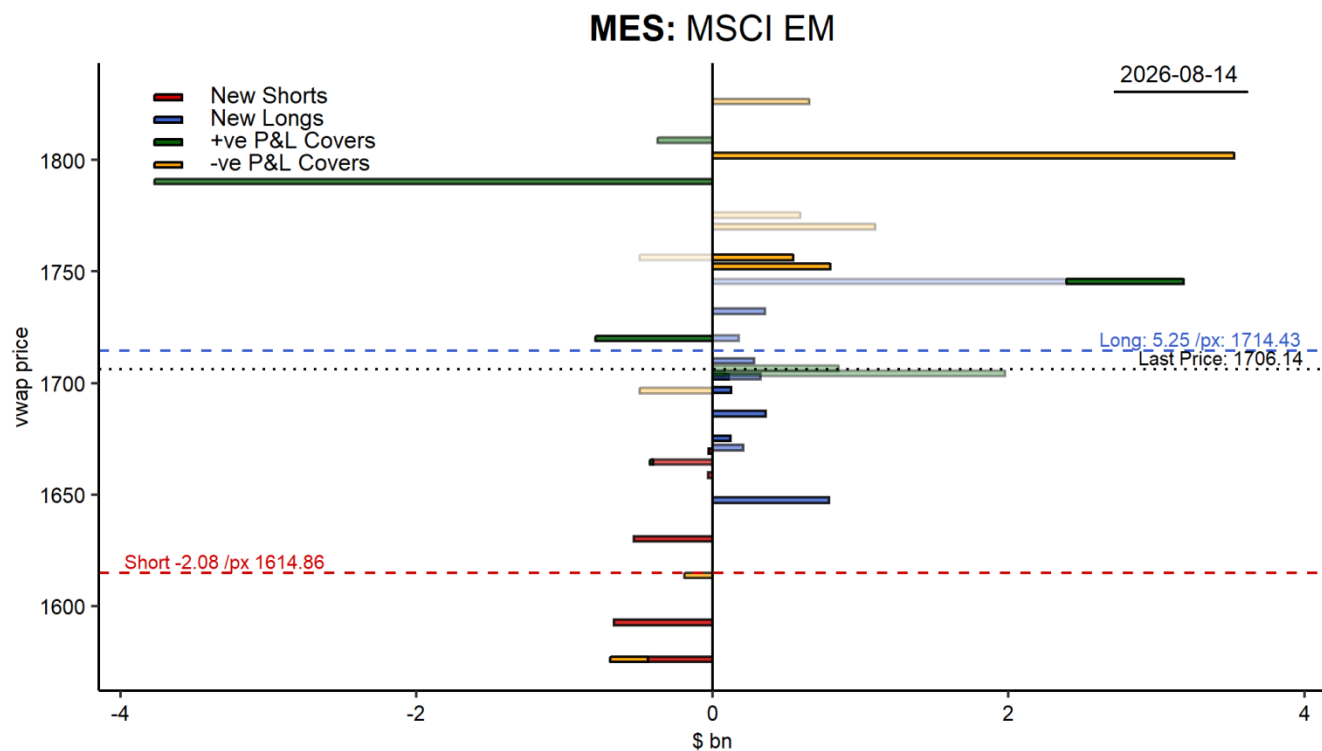


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

MSCI EM

Figure 27. MSCI EM Equity contracts (MES) – position snapshot and time-series

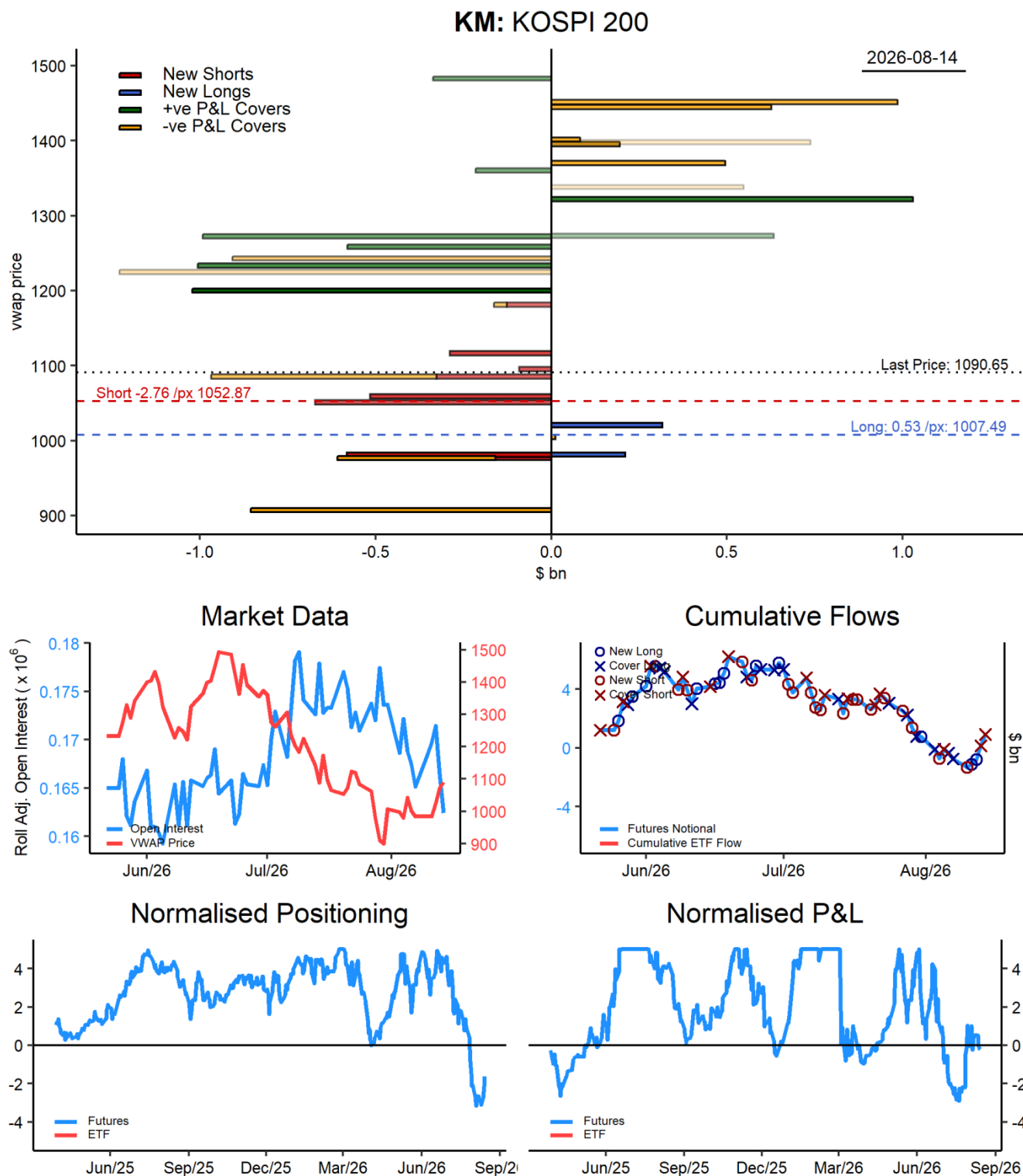


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

KOSPI 200

Figure 28. KOSPI 200 contracts (KM) – position snapshot and time-series

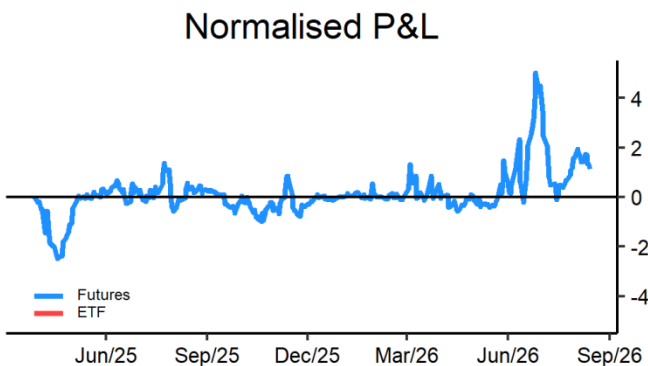
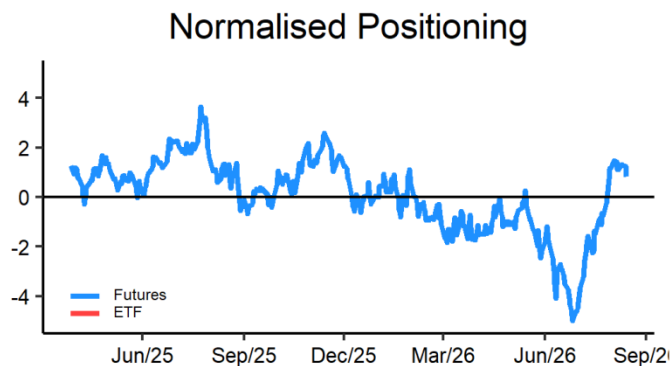
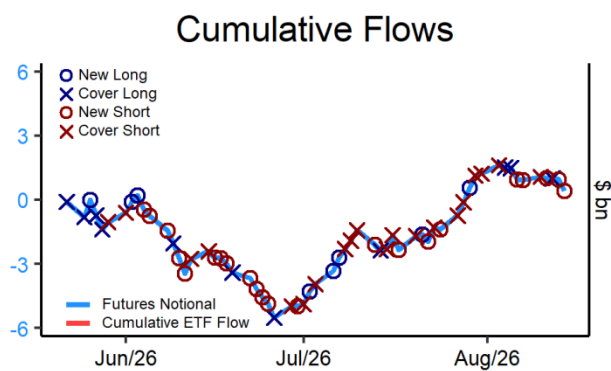
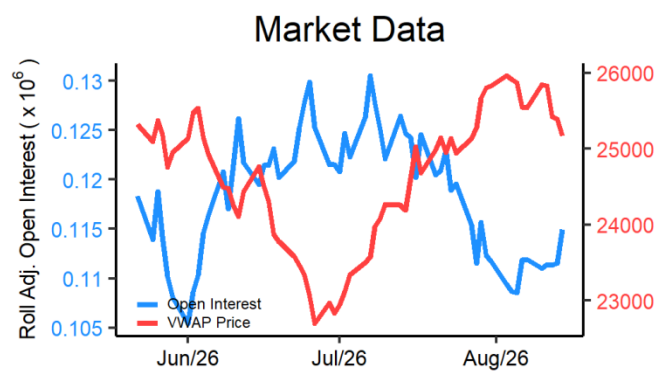
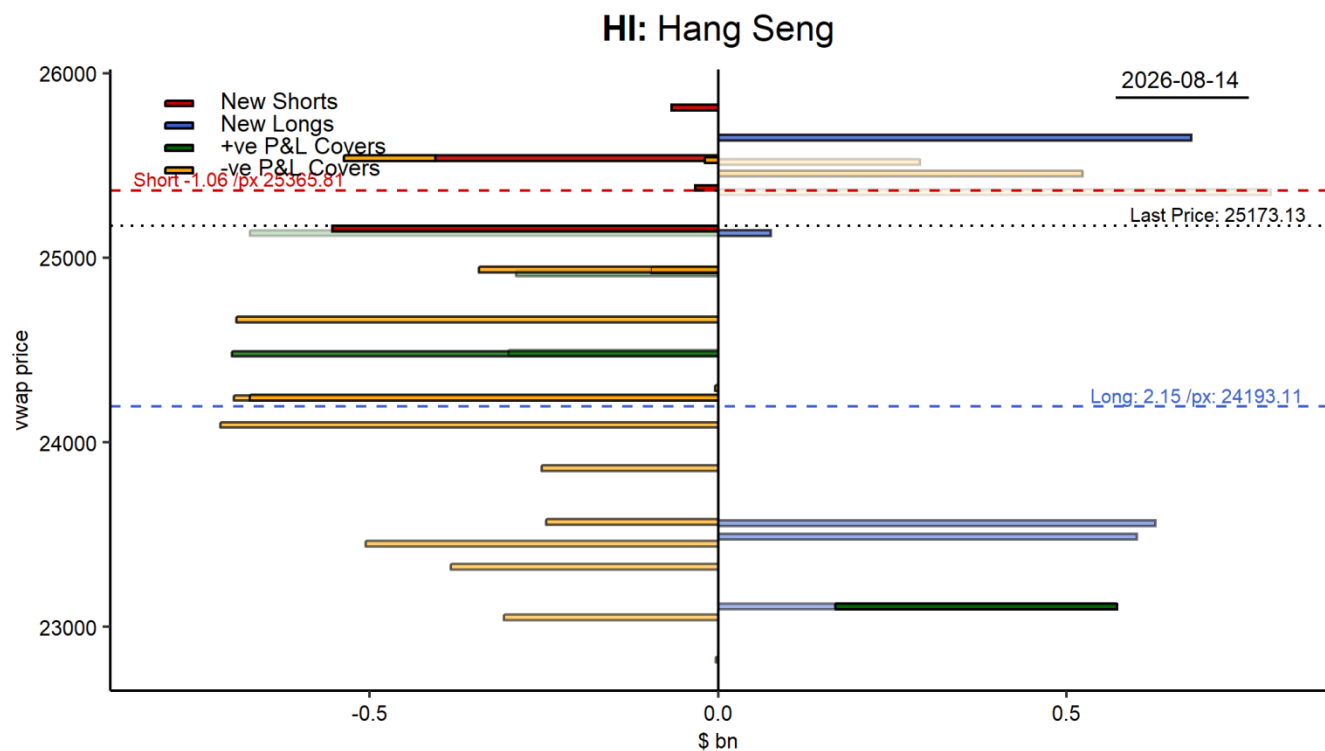


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

Hang Seng

Figure 29. Hang Seng contracts (HI) – position snapshot and time-series

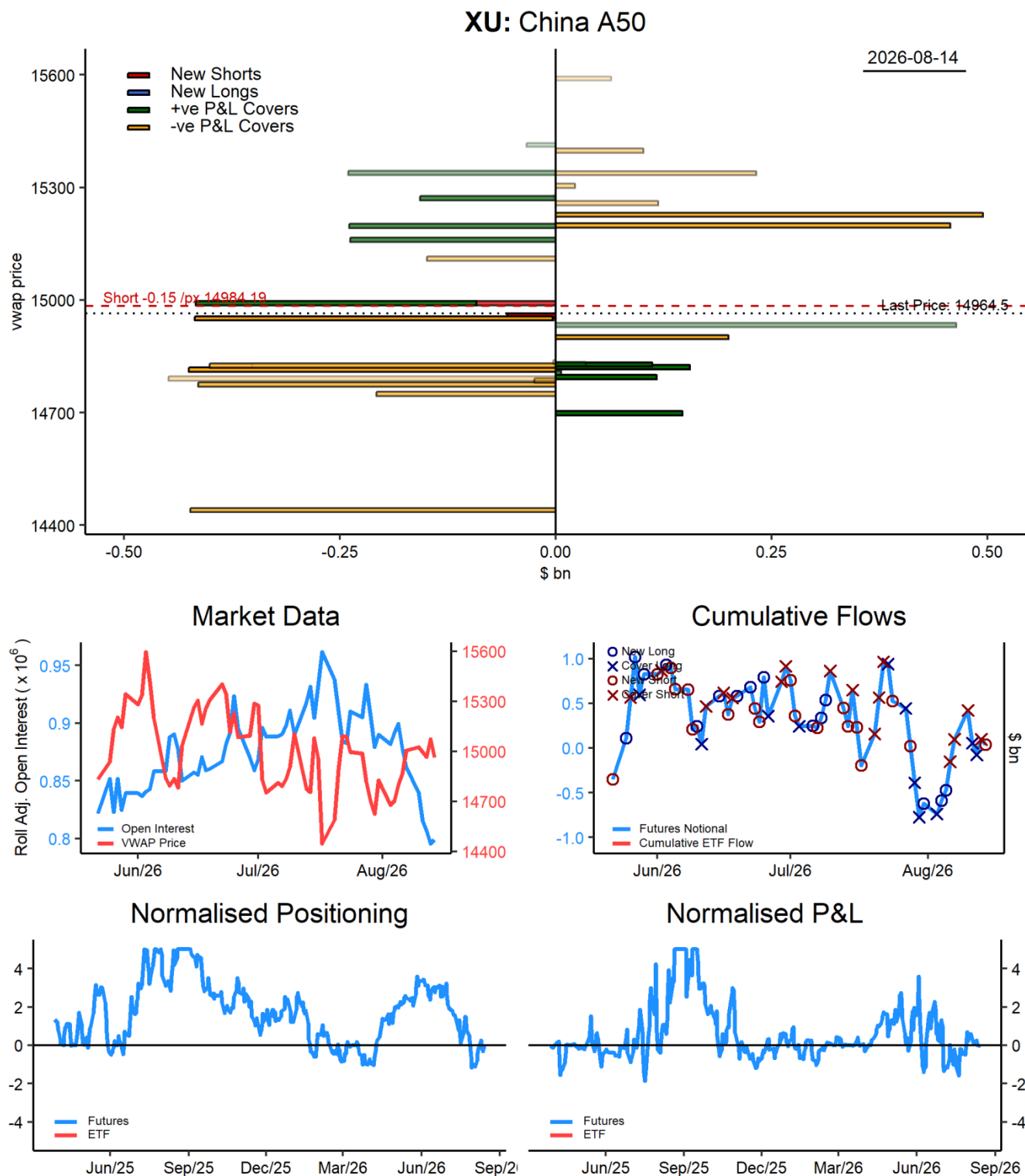


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

China A50

Figure 30. A50 contracts (XU) – position snapshot and time-series



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Bloomberg

The Equity Markets Positioning Model – Methodology

The aim of the Equity Markets Positioning Model (EMP) is to provide information on current equity market positioning and how this might influence subsequent price action. It allows readers to ‘visualize’ the investor exposures in equity futures across multiple markets in order to:

- Quantify the current market positioning and profitability;
- Identify extreme market positions;
- Categorize periods in which price action will be dominated by positioning;

The model calculates a ‘headline’ positioning score between -5 to 5 and provides a visualization of the current and past market.

How does the model work?

At a micro scale, price action is dominated by the size, level and speed at which investors choose to execute trades with other financial intermediaries. Buy/sell orders can clearly drive prices higher or lower and the EMP attempts to quantify these flows in order to understand the market direction and motivation.

This is a grand ambition, since there is no one market participant who holds a complete view of all market flows and we are only analyzing and aggregating futures trading to extrapolate the positioning across the whole market (i.e. including all cash trading). Based on this information we estimate:

- Market positioning: What is the outstanding market positioning (i.e. has there been net buying or selling recent history)? At what price and size were these trades executed?
- Market P&L: What is the unrealized profit or loss within the market (given the current outstanding market positioning)? When is this a motivation for investors to risk manage their positions?

We recognize that there are limits to this analysis – incomplete trade visibility, opaque instruments and unknown investor trading strategies create uncertainty in the ‘true’ picture of high-level equity market positioning.

The EMP model calculates the historical equity market positioning (over the last 3mths) and the associated unrealized P&L for the main markets around the world. A breakdown of the assets and contracts used in the EMP are presented in Figure 31 and calculation methodologies are outlined below.

Figure 31. Equity Markets covered by EMP Model

Country/Region	Index	Futures Ticker	ETF Flows
US	S&P 500	ES	Yes
US Tech	Nasdaq 100	NQ	Yes
US SMID	Russell 2000	RTY	No
Europe	Eurostoxx 50	VG	Yes
Germany	DAX 40	GX	Yes
U.K.	FTSE 100	Z	Yes
European Banks	SX7E	CA	Yes
Australia	S&P ASX 200	XP	No
Japan	Nikkei 225	NI	Yes
EAFE	MSCI ex-US	MFS	No
Emerging Markets	MSCI EM	MES	Yes
Korea	KOSPI 200	KM	No
Hong Kong	Hang Seng	HI	No
China	A50	XU	No

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

Positioning Calculations

The model identifies an objective subset of positions entered into (or closed) on a given day and classify these as being the relevant marginal long or short positions that determined (being the 'aggressor') the price move that day. In the report and in our analysis we will simply refer to these positions as longs, shorts and net position but they will always be a reference to this concept of a marginal positioning. This presumption then is that this subset of marginal positions are the most price sensitive, i.e. more likely to generate new flows (e.g. profit taking or squeezes) in response to future price moves.

In practice the model uses public trade activity data on the exchange traded futures contracts which is available daily. Naturally, net positioning is always zero for s, since the exchange matches buyers and sellers and creates and cancels contracts as necessary. However, by tracking the daily changes in overall open interest (the number of contracts after the exchange create or cancels contracts) and combining that measure with the change in price we can make a classification for the marginal positioning change on that day as set out in Figure 32.

The change in open interest is taken to be the relevant marginal subset of positions that day and together with the change in price we determine whether to consider it a change in long or short positions.

Figure 32. Positioning Matrix for Futures Markets

		Open Interest	
		Increasing	Decreasing
Daily Prices	Increasing	New Longs Rising prices and increasing open interest implies the 'aggressor' is the buyer pushing the market higher	Cover Shorts Rising prices and decreasing open interest implies the 'aggressor' is covering shorts (i.e. buying into the rally)
	Decreasing	New Shorts Falling prices and increasing open interest implies the 'aggressor' is the seller pushing the market lower	Cover Longs Falling prices and decreasing open interest implies the 'aggressor' is covering longs (i.e. selling into the sell-off)

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

Performing this analysis on a day-by-day basis creates a ‘trade activity’ history over a three-month period which can be converted into a snapshot of cumulative positioning and unrealized P&L. To illustrate the methodology, it is useful to look at a worked example. In Figure 33, we present these calculations for S&P 500 futures in early March 2020.

In this period, open interest increased and prices fell, suggesting an increase in the short investor base. The model shows numerous days where new short positions have been opened and relatively few where longs positions have been established, in addition to covering for some long and short positions. At the end of the period the cumulative uncovered positions and P&L is a simple the sum of the outstanding positions / P&L (see “total” in Figure 33 – cumulative positions equate to -459,041 contracts and 868mIn of unrealized P&L).

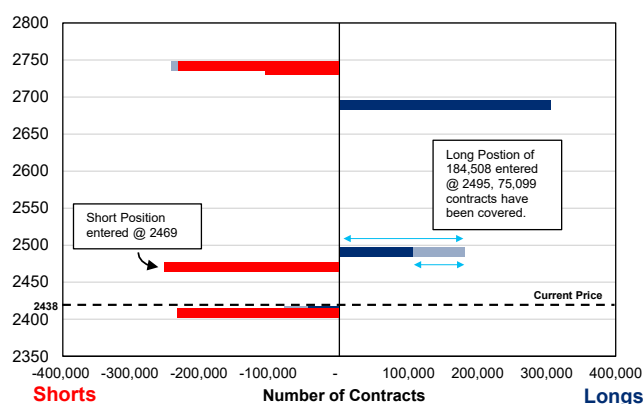
This positioning can be visualized as a horizontal bar graph ordered by price (see Figure 34). It can be clearly seen that the shorts were centered around 2740-2750 and the positions were significantly in the money. This contrasts with the big long position at 2696 which with a falling market was losing position. This snapshot of positioning can be summarized by the cumulative position unrealized P&L levels.

Figure 33. Example of positioning for S&P500 during March 2020

Date	Trade Price	Open Interest	Delta Price	Delta Interest	Trade Activity	Contract Buy/Sell	Cum. Pos.	P&L Snapshot (\$ mil)
06-Mar	2964.0	3,213,000						
09-Mar	2747.8	3,442,251	-7.3%	229,251	New Short	-229,251	-229,251	3,551
10-Mar	2865.8	3,438,501	4.3%	-3,750	Cover Short	3,750	-225,501	-58
11-Mar	2740.3	3,549,717	-4.4%	111,216	New Short	-111,216	-336,717	1,681
12-Mar	2469.0	3,808,784	-9.9%	259,067	New Short	-259,067	-595,784	402
13-Mar	2696.0	4,124,586	9.2%	315,802	New Long	315,802	-279,982	-4,074
16-Mar	2416.3	4,369,586	-10.4%	245,000	New Short	-245,000	-524,982	-266
17-Mar	2495.5	4,554,094	3.3%	184,508	New Long	184,508	-340,474	-531
18-Mar	2414.0	4,628,421	-3.3%	74,327	New Short	-74,327	-414,801	-89
19-Mar	2403.3	4,553,322	-0.4%	-75,099	Cover Long	-75,099	-489,900	216
20-Mar	2438.0	4,522,463	1.4%	-30,859	Cover Short	30,859	-459,041	37
Total							-459,041	868

© 2026 Citigroup Inc. No redistribution without Citigroup’s written permission.
Source: Citi Research

Figure 34. Example visualization of positioning



© 2026 Citigroup Inc. No redistribution without Citigroup’s written permission.
Source: Citi Research

In EMP, the daily volume weighted average price (VWAP) is used in the calculations (rather than closing prices), since it creates more robust results. In addition, the EMP model chooses to cover trades that have the maximum economic impact (i.e. trades with the largest gains or losses are closed out first). However, in every other aspect, the methodology is identical to the one presented above.

Finally, the reported numbers in the EMP model are normalised into a range from -5 to +5 (representing large shorts / losses to large longs / profits) in order to compare across asset / instruments.

What output is generated by EMP?

The EMP model provides a snapshot of the current positioning (see Figure 35 to Figure 37). These tables summarize the outstanding exposure and unrealized P&L (for a 3mth period in both futures and ETFs).

Typical output is shown in Figure 35 for cross equity market positioning in futures. In this illustration, investors are currently neutral to net short Eurostoxx 50 contracts, which has decreased from a net long position of the past 3 months. This is in contrast to MSCI EM futures where investors are sitting on profitable positions.

Figure 35. Example of Equity Market Positioning and PnL

All metrics in table are normalised between +5 and -5

Investors currently net long S&P 500 Emini futures

Investors currently neutral to net short EuroStoxx 50

(\$bn) 03/09/2020		S&P 500	DAX	FTSE 100	EStoxx50	Nasdaq	Nikkei	MSCI World x-US	MSCI EM
		ES	GX	Z	VG	NQ	NI	MFS	MES
Normalised Scores (-5 / +5)									
Positioning	Future	1.7	-0.1	1.0	-0.0	4.8	1.1	-0.7	2.0
	ETF	3.0	1.4	-0.6	0.9	3.6	1.0		1.1
	3m Net Pos	2.3	0.7	0.2	0.4	4.2	1.1	-0.7	1.6
	1w Change	0.4	0.2	0.3	0.3	-0.3	0.3	-0.4	-0.3
P&L	Future	1.2	-0.1	-0.7	-0.0	4.8	0.5	0.1	1.8
	ETF	3.7	1.9	-0.5	1.1	3.9	1.3		1.4
	3m Net PnL	2.5	0.9	-0.6	0.5	4.4	0.9	0.1	1.6
	1w Change	0.3	0.3	-0.0	0.3	-0.4	0.3	0.0	-1.1

Investors are sitting on profitable positions in MSCI EM.

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

In addition, we further breakdown market positioning by analyzing both long and short positions, the average entry price and the position relative to trading over the past three months (Figure 36).

Figure 36. Example of Breakdown of Market Positioning – Not Current

Net Notional = Longs minus Shorts (in \$blns)

Percentile refers to positioning relative to past 3 years

% Offside refers to what percentage of shorts are below Ref price – indicates risk of short squeeze if market rallies. Opposite for Longs

		S&P 500	DAX	FTSE 100	ESTOxx50	Nasdaq	Nikkei	MSCI World x-US	MSCI EM
(\$bn)	03/09/2020	ES	GX	Z	VG	NQ	NI	MFS	MES
Futures	Net Notional	31.7	-0.1	2.4	-0.4	10.8	1.2	-1.1	6.4
	Percentile	67%	36%	53%	33%	99%	64%	11%	75%
Shorts	Notional	-7.5	-1.3	-3.9	-1.4	-1.6	-1.1	-2.7	-0.7
	Percentile	24%	37%	81%	24%	28%	20%	80%	21%
	Avg. Entry	3,246	12,994	6,116	3,316	10,993	22,959	1,890	1,089
	% offside	100%	100%	3%	100%	100%	100%	32%	89%
Longs	Notional	39.2	1.2	6.3	0.9	12.4	2.3	1.6	7.0
	Percentile	57%	18%	72%	10%	99%	25%	18%	67%
	Avg. Entry	3,337	13,068	6,096	3,311	11,166	22,889	1,893	1,051
	% offside	13%	0%	100%	0%	18%	0%	60%	38%
	Ref Price	3,463	13,304	5,909	3,366	11,803	23,490	1,878	1,094
Activity	1-day	-2.6	-0.2	0.0	-0.1	-0.9	0.3	0.0	0.0
	Type	longs	shorts	longs	shorts	longs	longs	shorts	shorts
	1-week	4.2	0.1	0.1	-0.5	-0.4	0.1	-0.7	0.5

Average entry price over past 3 months

1 day trading activity in \$blns. Negative (positive) means risks is being cut (added)

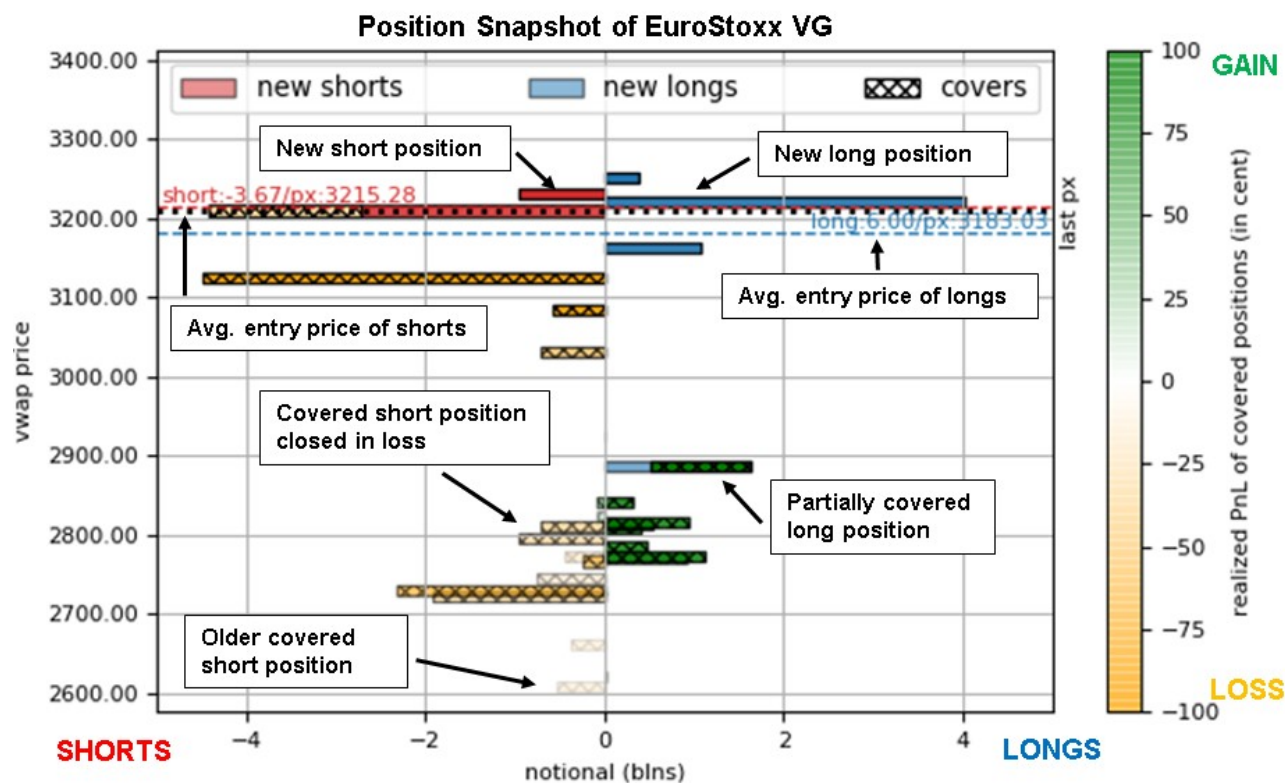
Defines what side the risk is being traded

1 week cumulative trading flow

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research

Lastly, a complete visualization of the positioning is presented in Figure 37. This shows the current positioning ordered by VWAP prices over a 3-month period. The bars in the chart illustrate the level at which trades were executed (red = short positions and blue = long positions). The lighter bars imply older trade activity. When a trade is covered the bar is hashed with XXX and the color of the shading represents if it was covered at a gain or loss (yellow=loss, green=gain). The horizontal red and blue dotted lines show the respective average entry price of contracts over the past three months.

Figure 37. Example Snapshot of Position in EuroStoxx Futures – Not Current



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

What it means for Cash Equities

Although traditionally used as a hedge, futures are also used by investors for taking an outright view on markets. This becomes increasingly relevant when futures positioning is extended. We explore this content in greater detail in [Regime Modelling using Futures Positioning: Futures vs. Cash – A Distant but Meaningful Relationship](#), which explores the interplay between futures positioning and behavior seen in underlying cash markets.

Previously Published Research

Figure 38. Research Reports (since 2009)

Date	Research Theme	Report Link
20-Oct-25	Allocation	Searching for Alpha: What can Mutual Fund Flows tell us about Future Stock and Sector Returns?
29-Sep-25	Factor	What Works in Equity Markets: Misguided by Factor Betas
04-Sep-25	Rotation	What Works in Equity Markets: Style Factor Performance and High Macro Risk Environments
01-May-25	Positioning	What Works in Equity Markets: Does Crowding Outperformance Outweigh Crash Risk?
23-Apr-25	Rotation	Searching for Alpha – Style Rotation: Using Mutual Fund Flows to Predict Style Returns
05-Dec-24	Factor	Searching for Alpha – Stock Selection and Enhanced Growth Strategies with Option-Implied Volatility
29-Oct-24	Allocation	Searching for Alpha – Macro Sector Selection – Using Macro Regimes & News Sentiment to Forecast Sector Returns
27-Aug-24	Positioning	Measuring the Crowded Trade in Global Equities News Sentiment, Factor Crowding & Extended Coverage
20-Aug-24	Misc.	Global Risk Attribute Model: A Framework for Equity Macro Risk Analysis
10-Apr-24	Misc	Searching for Alpha – European Industry-Specific Return Drivers
11-Jan-24	Allocation	Searching for Alpha – Are PMIs Leading or Misleading?
04-Dec-23	Factor	Searching for Alpha: Leveraging Option Information to Improve Factor Performance
26-Sep-23	Allocation	Searching for Alpha: Macro Regimes and Systematic Equity Investing
10-Aug-23	Allocation	Searching for Alpha: Exposure to Macro via Equities
26-Jun-23	Positioning	Measuring the Crowded Trade in Global Equities: Introducing our Short Stock Crowding Model
16-May-23	Machine Learning	Searching for Alpha: From Machine Learning to Stock Screens – A Rulefit Model
24-Mar-23	Factor	Searching for Alpha: A Bottom-Up Approach to Evaluating Market Overbought/Oversold Conditions
20-Jan-23	Factor	What Works in Equity Markets: The Enduring Appeal of Income Investing
13-Jan-23	Factor	What Works in Equity Markets: Return on Equity, a different perspective through Revisions
07-Nov-22	Event	Searching for Alpha: Systematic M&A Strategies that Outperform Discretionary Event-Driven Approaches
26-Oct-22	Allocation	What Works in Equity Markets: Sector Selection – A Bottom-up, Factor Approach
09-Jun-22	Event	Searching for Alpha: M&A – Identifying Potential Takeover Targets
24-Feb-22	Event	What Works in Equity Markets: Buy Backs – Back in Vogue
26-Jul-21	Factor	What Works in Equity Markets: Risky Value vs. Quality Value Revisited
26-Jul-21	Machine Learning	Searching for Alpha: Practical Considerations in Building Neural Networks for Stock Selection
22-Jul-21	ESG	ESG Insights: Indexed ESG Funds – A Widespread Implementation Choice
22-Jun-21	Factor	Searching for Alpha: Go with the Stock Connect Flow, for a Sharpe of 4
18-Jun-21	Factor	Searching for Alpha: Overlaying Price Mo. and Short Interest to avoid drawdowns and improve returns
15-Jun-21	ESG	ESG Insights: ESG Fundamental Characteristics
13-Apr-21	ESG	ESG Insights: Following the ESG Herd: Has ESG become a Crowded Trade?
29-Mar-21	Machine Learning	Searching for Alpha: Pairs Trading: Applying Machine Learning to Pairs Trading
09-Mar-21	Positioning	Regime Modelling using Futures Positioning: Futures vs. Cash – A Distant but Meaningful Relationship
18-Feb-21	ESG	ESG Insights: The ESG ‘Premium’: How does ESG compare to other Factors?
01-Feb-21	Event	Searching for Alpha: Asia Earnings Surprise: Predicting Asia Earnings Surprises
14-Jan-21	ESG	ESG Insights: Where Machines do it Better: Inferred ESG Ratings Data
30-Oct-20	Factor	Searching for Alpha: China A Alpha – Smoothing Price Momentum
16-Oct-20	Factor	Under the Microscope: Overlapping Momentum
14-Sep-20	Positioning	Equity Markets Positioning Model: Introducing the EMP
11-Sep-20	Factor	Searching for Alpha: Combining ESG and Risk Premia Investment: A Double Screen Approach
08-Sep-20	Factor	Searching for Alpha: Factoring Short Interest: Measuring and Profiting from Information in Shorts
28-May-20	Machine Learning	Searching for Alpha: Machine Learning – SHAP Value as Factor Selection Criterion
28-Feb-20	Machine Learning	Searching for Alpha: Machine Learning: Leveraging Return Forecasts
21-Feb-20	Factor	Searching for Alpha: China A Alpha – Sector-rel. Value Is Powerful, Choose Sector Mappings Wisely
03-Oct-19	Machine Learning	Searching for Alpha: Machine Learning: Beyond Random Forest for Stock Selection
23-Sep-19	Event	Searching for Alpha: Earnings Surprise: Managing Expectations
05-Mar-19	Machine Learning	Searching For Alpha: Machine Learning: Interpreting Machine Learning Predictions
07-Feb-19	Misc	Searching for Alpha: GAAP vs. Non-GAAP: Which Earnings does the Market Price?
21-Nov-18	Positioning	Searching for Alpha: Style Crowding in Asia: Getting Ahead of the Crowd
17-Sep-18	Event	Searching for Alpha: Earnings Surprise: Using ML to Forecast Earnings Surprises & Returns
13-Sep-18	ESG	Searching for Alpha: The ESG Edge: ESG Investing – A Step Forward
03-Sep-18	Machine Learning	Searching For Alpha: Machine Learning: Interacting Machine Learning and Factors
03-May-18	Positioning	Measuring the Crowded Trade: Introduction to our Crowding Composite for individual stocks
14-Mar-18	Factor	Searching for Alpha: Profiting from Capex: Look To Capex Announcements
07-Sep-17	Rotation	Searching for Alpha: Tactical Style Rotation: Using Risk and Return to Manage Style Exposure
10-Mar-17	Misc	Searching for Alpha: Big Data: Navigating New Alternative Datasets
13-Feb-17	Factor	Searching for Alpha: Betting Against (Accurate) Beta
19-Sep-16	Factor	Searching for Alpha: Competitive Advantage: Survival of the Fittest
09-Sep-16	Factor	Searching for Alpha: Financial Strength Redux
14-Apr-16	Rotation	Searching for Alpha: Dynamic Style Weighting: Risk-Based Equity Style Allocation
04-Mar-16	Smart Beta	Long-Only Pure Style Portfolios: No Shorts Please
18-Feb-16	Factor	Industry Alpha Insights: Banks: One Size Does Not Fit All
21-Sep-15	Factor	Under the Microscope: Stock Momentum Conflation
15-Sep-15	Allocation	Searching for Alpha: Macro Moves Markets: Economic Data, Expectations and Market Pricing
06-Aug-15	Misc	Normalising for Style Factor Composites: Reviewing alternative solutions for constructing style composites
23-Mar-15	Rotation	Searching for Alpha: Style Timing: Style Performance, Trading Volumes and Investor Agreement
13-Mar-15	Misc	World Radar Screen: Refining Our Global Search for Alpha
18-Feb-15	Factor	Searching for Alpha: Networking with Analysts: Modelling Analyst Forecast Dependence
01-Oct-14	Smart Beta	The Rise of Low Risk Investing: Is It Getting Crowded Out There?
27-Mar-14	Factor	Under the Microscope: Five Innovations In Momentum Investing
07-Mar-14	Factor	Searching for Alpha: Timing Price Momentum: When Does Momentum Move?
27-Nov-13	Smart Beta	Equity Risk Premia Investing: A New Methodology for Monitoring Style Performance
23-Jul-13	Allocation	Stock Market Country Selection: Changes to a Well-Established Model

02-Jul-13	Factor	Searching For Alpha: Digging For Dividends: QUARI - QUality with A Reliable Income
24-Jun-13	Misc	Global Theme Machine: An Objective Way of Identifying Attractive Investment Themes
25-Mar-13	Factor	Searching for Alpha: Purifying Analyst Recommendations: Removing Beta to get to the Alpha
06-Nov-12	Factor	Searching for Alpha: Tangible Benefits of Intangibles: Brand, Respect & Intellectual Capital
09-Mar-12	Smart Beta	Low-Risk Portfolio Strategies: Sharpe Ratio Maximisation and Multi-Asset Applications
28-Feb-12	Rotation	Macro Risk and Style Rotation: A Guide Rather than a Prescription
14-Sep-11	Factor	Searching for Alpha: Accruals Volatility - A New Approach to Quality Investing
24-Aug-11	Allocation	Industry Alpha Insights: Four Approaches to Tactical Industry Selection
17-Mar-11	Misc	Industry Alpha Insights: Quantifying Industry-Specific Fundamentals
18-Nov-10	Smart Beta	Low-Risk Equity Portfolios: More than just Minimum Variance
15-Nov-10	Allocation	Under the Microscope: Measuring Systemic Risk - The Absorption Ratio
14-Jun-10	Factor	Under the Microscope: Optionality in Valuation
31-Mar-10	Event	Searching for Alpha: Earnings Surprise: Still Profiting from Surprises
29-Jan-10	Factor	Momentum in Japan: Looking at Price, Trading Values and Earnings
15-Oct-09	Rotation	Searching for Alpha: Style Rotation: Optimising Style Rotation Strategies

Source: Citi Research

Citi Quant Research Team

Figure 39. Citi Quantitative Research Teams (For Informational Purposes Only)

Global Quantitative Research

Europe

David Chew¹ +44-20-7986-7698 david.chew@citi.com

North America

Anju Bhandari⁴ +1-212-816-3812 anju.bhandari@citi.com

Richard Schlatter⁴ +1-212-816 0591 richard.w.schlatter@citi.com

Asia

Yue Hin Pong¹ +44-20-7986-3953 yue.hin.pong@citi.com

Multi-Asset

Alex Saunders⁴ +1-212-723-1058 alexander.saunders@citi.com

¹ Citigroup Global Markets Ltd; ² Citigroup Global Markets Asia Limited; ³ Citigroup Pty Limited, ⁴ Citigroup Global Markets Inc.

RESEARCH ANALYST AFFILIATIONS: The Research Analyst Affiliations listed above (i.e., the research analysts listed above other than those identified as employed by Citigroup Global Markets inc.) are not registered/qualified as research analysts with FINRA. Such research analysts may not be associated persons of the member organization and therefore may not be subject to the NYSE FINRA 2241 restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account. Unless indicated in Appendix A-1 of this document or any of the referenced documents, the analysts listed above have not contributed to this document or any of the referenced documents.

If you are visually impaired and would like to speak to a Citi representative regarding the details of the graphics in this document, please call USA 1-888-500-5008 (TTY: 711), from outside the US +1-210-677-3788

Appendix A-1

ANALYST CERTIFICATION

The research analysts primarily responsible for the preparation and content of this research report are either (i) designated by "AC" in the author block or (ii) listed in bold alongside content which is attributable to that analyst. If multiple AC analysts are designated in the author block, each analyst is certifying with respect to the entire research report other than (a) content attributable to another AC certifying analyst listed in bold alongside the content and (b) views expressed solely with respect to a specific issuer which are attributable to another AC certifying analyst identified in the price charts or rating history tables for that issuer shown below. Each of these analysts certify, with respect to the sections of the report for which they are responsible: (1) that the views expressed therein accurately reflect their personal views about each issuer and security referenced and were prepared in an independent manner, including with respect to Citigroup Global Markets Inc. and its affiliates; and (2) no part of the research analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in this report.

IMPORTANT DISCLOSURES

Analysts' compensation is determined by Citi Research management and Citigroup's senior management and is based upon activities and services intended to benefit the investor clients of Citigroup Global Markets Inc. and its affiliates (the "Firm"). Compensation is not linked to specific transactions or recommendations. Like all Firm employees, analysts receive compensation that is impacted by overall Firm profitability which includes investment banking, sales and trading, and principal trading revenues. One factor in equity research analyst compensation is arranging corporate access events between institutional clients and the management teams of covered companies. Typically, company management is more likely to participate when the analyst has a positive view of the company.

For financial instruments recommended in the Product in which the Firm is not a market maker, the Firm is a liquidity provider in such financial instruments (and any underlying instruments) and may act as principal in connection with transactions in such instruments. The Firm is a regular issuer of traded financial instruments linked to securities that may have been recommended in the Product. The Firm regularly trades in the securities of the issuer(s) discussed in the Product. The Firm may engage in securities transactions in a manner inconsistent with the Product and, with respect to securities covered by the Product, will buy or sell from customers on a principal basis.

Unless stated otherwise neither the Research Analyst nor any member of their team has viewed the material operations of the Companies for which an investment view has been provided within the past 12 months.

For important disclosures (including copies of historical disclosures) regarding the companies that are the subject of this Citi Research product ("the Product"), please contact Citi Research, 388 Greenwich Street, 6th Floor, New York, NY, 10013, Attention: Legal/Compliance [E6WYB6412478]. In addition, the same important disclosures, with the exception of the Valuation and Risk assessments and historical disclosures, are contained on the Firm's disclosure website at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures. Valuation and Risk assessments can be found in the text of the most recent research note/report regarding the subject company. Pursuant to the Market Abuse Regulation a history of all Citi Research recommendations published during the preceding 12-month period can be accessed via Citi Velocity (<https://www.citivelocity.com/cv2>) or your standard distribution portal. Historical disclosures (for up to the past three years) will be provided upon request.

Citi Research Equity Ratings Distribution

Data current as of 01 Jul 2026	12 Month Rating			Catalyst Watch		
	Buy	Hold	Sell	Buy	Hold	Sell
Citi Research Global Fundamental Coverage (Neutral=Hold)	61%	32%	7%	36%	48%	16%
% of companies in each rating category that are investment banking clients	38%	42%	27%	42%	37%	36%
Citi Research Quantitative World Radar Screen Model Coverage	30%	40%	30%			
% of companies in each rating category that are investment banking clients	24%	16%	16%			
Citi Research Quantitative Latam Radar Screen Model Coverage	23%	57%	21%			
% of companies in each rating category that are investment banking clients	39%	40%	43%			
Citi Research Quantitative Asia Radar Screen Model Coverage	20%	60%	20%			
% of companies in each rating category that are investment banking clients	26%	10%	7%			
Citi Research Quantitative Australia Radar Screen Model Coverage	41%	0%	59%			

% of companies in each rating category that are investment banking clients	35%	0%	4%
--	-----	----	----

Guide to Citi Research Fundamental Research Investment Ratings:

Citi Research stock recommendations include an investment rating and an optional risk rating to highlight high risk stocks. **Risk rating** takes into account both price volatility and fundamental criteria. Stocks will either have no risk rating or a High risk rating assigned.

Investment Ratings: Citi Research investment ratings are Buy, Neutral and Sell. Our ratings are a function of analyst expectations of expected total return ("ETR") and risk. ETR is the sum of the forecast price appreciation (or depreciation) plus the dividend yield for a stock within the next 12 months. The target price is based on a 12 month time horizon. The Investment rating definitions are: Buy (1) ETR of 15% or more or 25% or more for High risk stocks; and Sell (3) for negative ETR. Any covered stock not assigned a Buy or a Sell is a Neutral (2). For stocks rated Neutral (2), if an analyst believes that there are insufficient valuation drivers and/or investment catalysts to derive a positive or negative investment view, they may elect with the approval of Citi Research management not to assign a target price and, thus, not derive an ETR. Citi Research may suspend its rating and target price and assign "Rating Suspended" status for regulatory and/or internal policy reasons. Citi Research may also suspend its rating and target price and assign "Under Review" status for other exceptional circumstances (e.g. lack of information critical to the analyst's thesis, trading suspension) affecting the company and/or trading in the company's securities. In both such situations, the rating and target price will show as "—" and "—" respectively in the rating history price chart. Prior to 11 April 2022 Citi Research assigned "Under Review" status to both situations and prior to 11 Nov 2020 only in exceptional circumstances. As soon as practically possible, the analyst will publish a note re-establishing a rating and investment thesis. Investment ratings are determined by the ranges described above at the time of initiation of coverage, a change in investment and/or risk rating, or a change in target price (subject to limited management discretion). At times, the expected total returns may fall outside of these ranges because of market price movements and/or other short-term volatility or trading patterns. Such interim deviations will be permitted but will become subject to review by Research Management. Your decision to buy or sell a security should be based upon your personal investment objectives and should be made only after evaluating the stock's expected performance and risk.

Catalyst Watch/Short Term Views ("STV") Ratings Disclosure:

Catalyst Watch and STV Upside/Downside calls: Citi Research may also include a Catalyst Watch or STV Upside or Downside call to indicate the analyst expects the share price to rise (fall) in absolute terms over a specified period of 30 or 90 days in reaction to one or more specific near-term catalysts or events impacting the company or the market. A Catalyst Watch will be published when Analyst confidence is high that an impact to share price will occur; it will be a STV when confidence level is moderate. A Catalyst Watch or STV Upside/Downside call will automatically expire at the end of the specified 30/90 day period. The Catalyst Watch will also be automatically removed if share price performance (calculated at market close) exceeds 15% against the direction of the call (unless over-ridden by the analyst). The analyst may also remove a Catalyst Watch or STV call prior to the end of the specified period in a published research note. A Catalyst Watch/STV Upside or Downside call may be different from and does not affect a stock's fundamental equity rating, which reflects a longer-term total absolute return expectation. For purposes of FINRA ratings-distribution-disclosure rules, a Catalyst Watch/STV Upside call corresponds to a buy recommendation and a Catalyst Watch/STV Downside call corresponds to a sell recommendation. Any stock not assigned to a Catalyst Watch Upside, Catalyst Watch Downside, STV Upside, or STV Downside call is considered Catalyst Watch/STV No View. For purposes of FINRA ratings distribution-disclosure rules, we correspond Catalyst Watch/STV No View to Hold in our ratings distribution table for our Catalyst Watch/STV Upside/Downside rating system. However, we reiterate that we do not consider No View to be a recommendation. For all Catalyst Watch/STV Upside/Downside calls, risk exists that the catalyst(s) and associated share-price movement will not materialize as expected.

Privately held companies:

The legal entities through which Citi conducts its business do not permit investment research recommendations, ratings, target prices, or estimates on any privately held companies ("private companies") referenced herein. Citi Research does not utilize a rating system for private companies. Any information on private companies in this report is based on generally available information and, although obtained from sources believed by Citi to be reliable, its accuracy and completeness are not guaranteed. Private companies are generally not subject to reporting obligations, which may result in less publicly available information. The information contained in this report may differ from information a private company may subsequently make public or otherwise share with investors. Continuation of publishing private company reports or private company sections within reports is at Citi's discretion, in accordance with internal policies.

Guide to Citi Research Quantitative Research Investment Ratings:

Citi Research Quantitative Research World Radar Screen recommendations are based on a globally consistent framework to measure relative value and momentum for a large number of stocks across global developed and emerging markets. Relative value and momentum rankings are equally weighted to produce a global attractiveness score for each stock. The scores are then ranked and put into deciles. A stock with a decile rating of 1 denotes an attractiveness score in the top 10% of the universe (most attractive). A stock with a decile rating of 10 denotes an attractiveness score in the bottom 10% of the universe (least attractive).

Citi Research Asia Quantitative Radar Screen model recommendations are based on a regionally consistent framework to measure relative value and momentum for a large number of stocks across regional developed and emerging markets. Relative value and momentum rankings are equally weighted to produce a global attractiveness score for each stock. The scores are then ranked and put into quintiles. A stock with a quintile rating of 1 denotes an attractiveness score in the top 20% of the

universe (most attractive). A stock with a quintile rating of 5 denotes an attractiveness score in the bottom 20% of the universe (least attractive).

Citi Research Quantitative Research Latam Radar Screen recommendations are based on a globally consistent framework to measure relative value and momentum for a large number of stocks across markets in Latin America. Relative value and momentum rankings are equally weighted to produce a global attractiveness score for each stock. The scores are then ranked and put into quintiles. A stock with a quintile rating of 1 denotes an attractiveness score in the top 20% of the universe (most attractive). A stock with a quintile rating of 5 denotes an attractiveness score in the bottom 20% of the universe (least attractive).

Citi Research Australia Quantitative Radar Screen model recommendations are based on a robust framework to measure relative value and momentum for a large number of stocks across the Australian market. Stocks with a ranking of 1 denotes a stock that is above average in terms of both value and momentum relative to the stocks in the Australian market. A ranking of 10 denotes a stock that is below average in terms of both value and momentum relative to the stocks in the Australian market.

For purposes of FINRA ratings-distribution-disclosure rules, a Citi Research Quantitative World Radar Screen recommendation of (1), (2) or (3) most closely corresponds to a buy recommendation; a recommendation from this product group of (4), (5), (6) or (7) most closely corresponds to a hold recommendation; and a recommendation of (8), (9) or (10) most closely corresponds to a sell recommendation. An (NR) recommendation indicates that the stock is no longer in the screen. For purposes of FINRA ratings-distribution-disclosure rules, a Citi Research Latam Radar Screen recommendation of (1) most closely corresponds to a buy recommendation; a recommendation of (2),(3) or (4) most closely corresponds to a hold recommendation; and a recommendation of (5) most closely corresponds to a sell recommendation. An (NR) recommendation indicates that the stock is no longer in the screen.

For purposes of FINRA ratings distribution disclosure rules, a Citi Research Asia Quantitative Radar Screen recommendation of (1) most closely corresponds to a buy recommendation; a Citi Research Asia Quantitative Radar Screen recommendation of (2), (3), (4) most closely corresponds to a hold recommendation; and a recommendation of (5) most closely corresponds to a sell recommendation. An (NR) recommendation indicates that the stock is no longer in the screen.

For purposes of FINRA ratings-distribution-disclosure rules, a Citi Research Quantitative Research Australia Radar Screen recommendation of "attractive" (1) most closely corresponds to a buy recommendation. All other stocks in the sector are considered to be "unattractive" (10) which most closely corresponds to a sell recommendation. An (NR)/(O) recommendation indicates that the stock is no longer in the screen.

Recommendations are based on the relative attractiveness of a stock, thus can not be directly equated to buy, hold and sell categories. Accordingly, your decision to buy or sell a security should be based on your personal investment objectives and only after evaluating the stock's expected relative performance.

RESEARCH ANALYST AFFILIATIONS / NON-US RESEARCH ANALYST DISCLOSURES

The legal entities employing the authors of this report are listed below (and their regulators are listed further herein). Non-US research analysts who have prepared this report (i.e., all research analysts listed below other than those identified as employed by Citigroup Global Markets Inc.) are not registered/qualified as research analysts with FINRA. Such research analysts may not be associated persons of the member organization (but are employed by an affiliate of the member organization) and therefore may not be subject to the FINRA Rule 2241 restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Citigroup Global Markets Inc.

Richard W Schlatter; Anju Bhandari

Citigroup Global Markets Limited

Yue Hin Pong; David T Chew

OTHER DISCLOSURES

Any price(s) of instruments mentioned in recommendations are as of the prior day's market close on the primary market for the instrument, unless otherwise stated.

The completion and first dissemination of any recommendations made within this research report are as of the Eastern date-time displayed at the top of the Product. If the Product references views of other analysts then please refer to the price chart or rating history table for the date/time of completion and first dissemination with respect to that view.

Regulations in various jurisdictions require that where a recommendation differs from any of the author's previous recommendations concerning the same financial instrument or issuer that has been published during the preceding 12-month period that the change(s) and the date of that previous recommendation are indicated. For fundamental coverage please refer to the price chart or rating change history within this disclosure appendix or the issuer disclosure summary at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures.

Citi Research has implemented policies for identifying, considering and managing potential conflicts of interest arising as a result of publication or distribution of investment research. A description of these policies can be found at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures.

The proportion of all Citi Research research recommendations that were the equivalent to "Buy", "Hold", "Sell" at the end of each quarter over the prior 12 months (with the % of these that had received investment firm services from Citi in the prior 12 months shown in brackets) is as follows; Q1 2026 Buy 33%(63%), Hold 44% (52%), Sell 23% (46%), RV 0.5% (89%); Q4 2025 Buy 33% (63%), Hold 44% (50%), Sell 23% (46%), RV 0.4% (91%); Q3 2025 Buy 33% (61%), Hold 44% (52%), Sell 23% (50%), RV 0.4% (80%); Q2 2025 Buy 33%(63%), Hold 44% (51%), Sell 23% (49%), RV 0.4% (86%). For the purposes of disclosing recommendations other than for equity (whose definitions can be found in the corresponding disclosure sections), "Buy"

means a positive directional trade idea; "Sell" means a negative directional trade idea; and "Relative Value" means any trade idea which does not have a clear direction to the investment strategy.

European regulations require a 5 year price history when past performance of a security is referenced. CitiVelocity's Charting Tool (https://www.citivelocity.com/cv2/#go/CHARTING_3_Equities) provides the facility to create customisable price charts including a five year option. This tool can be found in the Data & Analytics section under any of the asset class menus in CitiVelocity (<https://www.citivelocity.com/>). For further information contact CitiVelocity support (https://www.citivelocity.com/cv2/go/CLIENT_SUPPORT). The source for all referenced prices, unless otherwise stated, is DataCentral, which sources price information from LSEG Data & Analytics. Past performance is not a guarantee or reliable indicator of future results. Forecasts are not a guarantee or reliable indicator of future performance.

Investors should always consider the investment objectives, risks, and charges and expenses of an ETF carefully before investing. The applicable prospectus and key investor information document (as applicable) for an ETF should contain this and other information about such ETF. It is important to read carefully any such prospectus before investing. Clients may obtain prospectuses and key investor information documents for ETFs from the applicable distributor or authorized participant, the exchange upon which an ETF is listed and/or from the applicable website of the applicable ETF issuer. The value of the investments and any accruing income may fall or rise. Any past performance, prediction or forecast is not indicative of future or likely performance. Any information on ETFs contained herein is provided strictly for illustrative purposes and should not be deemed an offer to sell or a solicitation of an offer to purchase units of any ETF either explicitly or implicitly. The opinions expressed are those of the authors and do not necessarily reflect the views of ETF issuers, any of their agents or their affiliates.

Citigroup Global Markets India Private Limited and/or its affiliates may have, from time to time, actual or beneficial ownership of 1% or more in the debt securities of the subject issuer.

Please be advised that pursuant to Executive Order 13959 as amended (the "Order"), U.S. persons are prohibited from investing in securities of any company determined by the United States Government to be the subject of the Order. This research is not intended to be used or relied upon in any way that could result in a violation of the Order. Investors are encouraged to rely upon their own legal counsel for advice on compliance with the Order and other economic sanctions programs administered and enforced by the Office of Foreign Assets Control of the U.S. Treasury Department.

This communication is directed at persons who are "Eligible Clients" as such term is defined in the Israeli Regulation of Investment Advice, Investment Marketing and Investment Portfolio Management law, 1995 (the "Advisory Law"). Within Israel, this communication is not intended for retail clients and Citi will not make such products or transactions available to retail clients or to non-Eligible Clients. The presenter is not licensed as investment advisor or investment marketer by the Israeli Securities Authority ("ISA") and this communication does not constitute investment or marketing advice. The information contained herein may relate to matters that are not regulated by the ISA. Any securities which are the subject of this communication may not be offered or sold to any Israeli person except pursuant to a security offering exemption according to the Israeli Securities Law, 1968 and the public offering rules provided thereunder.

Citi Research broadly and simultaneously disseminates its research content to the Firm's institutional and retail clients via the Firm's proprietary electronic distribution platforms (e.g., Citi Velocity and various Global Wealth platforms). As a convenience, certain, but not all, research content may be distributed through third party aggregators. Clients may receive published research reports by email, on a discretionary basis, and only after such research content has been broadly disseminated. Certain research is made available only to institutional investors to satisfy regulatory requirements. The level and types of services provided by Citi Research analysts to clients may vary depending on various factors such as the client's individual preferences as to the frequency and manner of receiving communications from analysts, the client's risk profile and investment focus and perspective (e.g. market-wide, sector specific, long term, short-term etc.), the size and scope of the overall client relationship with the Firm and legal and regulatory constraints.

Pursuant to Comissão de Valores Mobiliários Resolução 20 and ASIC Regulatory Guide 264, Citi is required to disclose whether a Citi related company or business has a commercial relationship with the subject company. Considering that Citi operates multiple businesses in more than 100 countries around the world, it is likely that Citi has a commercial relationship with the subject company.

Securities recommended, offered, or sold by the Firm: (i) are not insured by the Federal Deposit Insurance Corporation; (ii) are not deposits or other obligations of any insured depository institution (including Citibank); and (iii) are subject to investment risks, including the possible loss of the principal amount invested. The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security. Any decision to purchase securities mentioned in the Product must take into account existing public information on such security or any registered prospectus. Although information has been obtained from and is based upon sources that the Firm believes to be reliable, we do not guarantee its accuracy and it may be incomplete and condensed. Note, however, that the Firm has taken all reasonable steps to determine the accuracy and completeness of the disclosures made in the Important Disclosures section of the Product. The Firm's research department has received assistance from the subject company(ies) referred to in this Product including, but not limited to, discussions with management of the subject company(ies). Firm policy prohibits research analysts from sending draft research to subject companies. However, it should be presumed that the author of the Product has had discussions with the subject company to ensure factual accuracy prior to publication. Statements and views concerning ESG (environmental, social, governance) factors are typically based upon public statements made by the affected company or other public news, which the author may not have independently verified. ESG factors are one consideration that investors may choose to examine when making investment decisions. All opinions, projections and estimates constitute the judgment of the author as of the date of the Product and these, plus any other information contained in the Product, are subject to change without

notice. Prices and availability of financial instruments also are subject to change without notice. Notwithstanding other departments within the Firm advising the companies discussed in this Product, information obtained in such role is not used in the preparation of the Product. Although Citi Research does not set a predetermined frequency for publication, if the Product is a fundamental equity or credit research report, it is the intention of Citi Research to provide research coverage of the covered issuers, including in response to news affecting the issuer. For non-fundamental research reports, Citi Research may not provide regular updates to the views, recommendations and facts included in the reports. Notwithstanding that Citi Research maintains coverage on, makes recommendations concerning or discusses issuers, Citi Research may be periodically restricted from referencing certain issuers due to legal or policy reasons. Where a component of a published trade idea is subject to a restriction, the trade idea will be removed from any list of open trade ideas included in the Product. Upon the lifting of the restriction, the trade idea will either be re-instated in the open trade ideas list if the analyst continues to support it or it will be officially closed. Citi Research may provide different research products and services to different classes of customers (for example, based upon long-term or short-term investment horizons) that may lead to differing conclusions or recommendations that could impact the price of a security contrary to the recommendations in the alternative research product, provided that each is consistent with the rating system for each respective product.

Investing in non-U.S. securities, including ADRs, may entail certain risks. The securities of non-U.S. issuers may not be registered with, nor be subject to the reporting requirements of the U.S. Securities and Exchange Commission. There may be limited information available on foreign securities. Foreign companies are generally not subject to uniform audit and reporting standards, practices and requirements comparable to those in the U.S. Securities of some foreign companies may be less liquid and their prices more volatile than securities of comparable U.S. companies. In addition, exchange rate movements may have an adverse effect on the value of an investment in a foreign stock and its corresponding dividend payment for U.S. investors. Net dividends to ADR investors are estimated, using withholding tax rates conventions, deemed accurate, but investors are urged to consult their tax advisor for exact dividend computations. Investors who have received the Product from the Firm may be prohibited in certain states or other jurisdictions from purchasing securities mentioned in the Product from the Firm. Please ask your Financial Consultant for additional details. Citigroup Global Markets Inc. takes responsibility for the Product in the United States. Any orders by US investors resulting from the information contained in the Product may be placed only through Citigroup Global Markets Inc.

The Citigroup legal entity that takes responsibility for the production of the Product is the legal entity which the first named author is employed by.

The Product is made available in **Australia** through Citigroup Global Markets Australia Pty Limited. (ABN 64 003 114 832 and AFSL No. 240992), participant of the ASX Group and regulated by the Australian Securities & Investments Commission. Citigroup Centre, 2 Park Street, Sydney, NSW 2000. Citigroup Global Markets Australia Pty Limited is not an Authorised Deposit-Taking Institution under the Banking Act 1959, nor is it regulated by the Australian Prudential Regulation Authority.

The Product is made available in **Brazil** by Citigroup Global Markets Brasil - CCTVM SA, which is regulated by CVM - Comissão de Valores Mobiliários ("CVM"), BACEN - Brazilian Central Bank, APIMEC - Associação dos Analistas e Profissionais de Investimento do Mercado de Capitais and ANBIMA - Associação Brasileira das Entidades dos Mercados Financeiro e de Capitais. Av. Paulista, 1111 - 14º andar(parte) - CEP: 01311920 - São Paulo - SP.

This Product is available in **Chile** through Banchile Corredores de Bolsa S.A., an indirect subsidiary of Citigroup Inc., which is regulated by the Comisión Para El Mercado Financiero. Enrique Foster Sur, 20, piso 6, Las Condes, Santiago, Chile.

Disclosure for investors in the Republic of Colombia : This communication or message does not constitute a professional recommendation to make investment in the terms of article 2.40.1.1.2 of Decree 2555 de 2010 or the regulations that modify, substitute or complement it. Para la elaboración y distribución de informes de investigación y de comunicaciones generales de que trata este artículo no se requiere ser una entidad vigilada por la Financiera de Colombia.

The Product is made available in **Germany** by Citigroup Global Markets Europe AG ("CGME"), which is regulated by the European Central Bank and the German Federal Financial Supervisory Authority (Bundesanstalt für den Aufsicht über Kreditinstitute und Finanzdienstleistungsinstitute, BaFin). Börsenplatz 9, 60313 Frankfurt am Main, Germany.

Unless otherwise specified, if the analyst who prepared this report is based in Hong Kong and it relates to "securities" (as defined in the Securities and Futures Ordinance (Cap.571 of the Laws of Hong Kong)), the report is issued in Hong Kong by Citigroup Global Markets Asia Limited. Citigroup Global Markets Asia Limited is regulated by Hong Kong Securities and Futures Commission. If the report is prepared by a non-Hong Kong based analyst, please note that such analyst (and the legal entity that the analyst is employed by or accredited to) is not licensed/registered in Hong Kong and they do not hold themselves out as such. Please refer to the section "Research Analyst Affiliations / Non-US Research Analyst Disclosures" for the details of the employment entity of the analysts.

The Product is made available in **India** by Citigroup Global Markets India Private Limited (CGM), which is regulated by the Securities and Exchange Board of India (SEBI), as a Research Analyst (SEBI Registration No. INH000000438). CGM is also actively involved in the business of merchant banking (SEBI Registration No. INM000010718) and stock brokerage ((SEBI Registration No. INZ000263033) in India, and is registered with SEBI in this regard. Registration granted by SEBI, enlistment with Bombay Stock Exchange (BSE) and certification from National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. CGM's registered office is at 1202, 12th Floor, First International Financial Centre (FIFC), G Block, Bandra Kurla Complex, Bandra East, Mumbai - 400098 & registered Tel: +91 22 61759999. Citi maintains robust policies, procedures, controls, and training to ensure continued compliance with all applicable rules and regulations. All recommendations contained herein are made by duly qualified research analysts. CGM's Corporate Identity Number is 26657, and its Compliance Officer [Vishal Bohra] contact details are:

Tel: +91-022-61759994, Fax: +91-022-61759851, Email: cgmcompliance@citi.com. The Investor Charter in respect of Research Analysts, the Compliance Audit Report and Complaints information can be found at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures. The grievance officer [Niraj Mody] contact details are Tel: +91-22-42775002, Email: EMEA.CR.Complaints@citi.com. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. SEBI prescribed Client Terms & Conditions can be found at https://www.citivelocity.com/rendition/_/eppublic/V1/file?paramData=L0dETVMvcHVlVXNlLmhObWw

The Product is made available in **Indonesia** through PT Citigroup Sekuritas Indonesia. Citibank Tower 10/F, Pacific Century Place, SCBD lot 10, Jl. Jend Sudirman Kav 52-53, Jakarta 12190, Indonesia. Neither this Product nor any copy hereof may be distributed in Indonesia or to any Indonesian citizens wherever they are domiciled or to Indonesian residents except in compliance with applicable capital market laws and regulations. This Product is not an offer of securities in Indonesia. The securities referred to in this Product have not been registered with the Capital Market and Financial Services Authority (OJK) pursuant to relevant capital market laws and regulations, and may not be offered or sold within the territory of the Republic of Indonesia or to Indonesian citizens through a public offering or in circumstances which constitute an offer within the meaning of the Indonesian capital market laws and regulations.

The Product is made available in **Japan** by Citigroup Global Markets Japan Inc. ("CGMJ"), which is regulated by Financial Services Agency, Securities and Exchange Surveillance Commission, Japan Securities Dealers Association, Tokyo Stock Exchange and Osaka Securities Exchange. Otemachi Park Building, 1-1-1 Otemachi, Chiyoda-ku, Tokyo 100-8132 Japan. In the event that an error is found in an CGMJ research report, a revised version will be posted on the Firm's Citi Velocity website. If you have questions regarding Citi Velocity, please call (81 3) 6270-3019 for help.

The product is made available in the **Kingdom of Saudi Arabia** in accordance with Saudi laws through Citigroup Saudi Arabia, which is regulated by the Capital Market Authority (CMA) under CMA license (17184-31). 2239 Al Urubah Rd – Al Olaya Dist. Unit No. 18, Riyadh 12214 – 9597, Kingdom Of Saudi Arabia.

The Product is made available in **Korea** by Citigroup Global Markets Korea Securities Ltd. (CGMK), which is regulated by the Financial Services Commission, the Financial Supervisory Service and the Korea Financial Investment Association (KOFIA). The address of CGMK is Citibank Center, 50 Saemunan-ro, Jongno-gu, Seoul 03184, Korea. KOFIA makes available registration information of research analysts on its website. Please visit the following website if you wish to find KOFIA registration information on research analysts of

CGMK. <http://dis.kofia.or.kr/websquare/index.jsp?w2xPath=/wq/fundMgr/tList.xml&divisionId=MDISO3002002000000&serviceld=SDISO3002002000>. The Product is made available in Korea by Citibank Korea Inc., which is regulated by the Financial Services Commission and the Financial Supervisory Service. Address is Citibank Center, 50 Saemunan-ro, Jongno-gu, Seoul 03184, Korea. This research report is intended to be provided only to Professional Investors as defined in the Financial Investment Services and Capital Market Act and its Enforcement Decree in Korea.

The Product is made available in **Malaysia** by Citigroup Global Markets Malaysia Sdn Bhd (Registration No. 199801004692 (460819-D)) ("CGMM") to its clients and CGMM takes responsibility for its contents as regards CGMM's clients. CGMM is regulated by the Securities Commission Malaysia. Please contact CGMM at Level 43 Menara Citibank, 165 Jalan Ampang, 50450 Kuala Lumpur, Malaysia in respect of any matters arising from, or in connection with, the Product.

The Product is made available in **Mexico** by Citi México Casa de Bolsa, S.A. de C.V., Grupo Financiero Citi México which is a wholly owned subsidiary of Citigroup Inc. and is regulated by Comisión Nacional Bancaria y de Valores. Prolongación Reforma 1196, 24 floor, Colonia Santa Fe, Alcaldía Cuajimalpa de Morelos, C.P. 05348, Ciudad de México.

The Product is made available in **Poland** by Biuro Maklerskie Banku Handlowego (DMBH), separate department of Bank Handlowy w Warszawie S.A. a subsidiary of Citigroup Inc., which is regulated by Komisja Nadzoru Finansowego. Biuro Maklerskie Banku Handlowego (DMBH), ul. Senatorska 16, 00-923 Warszawa.

The Product is made available in **Singapore** through Citigroup Global Markets Singapore Pte. Ltd. ("CGMSPL"), a capital markets services and Exempt Financial Advisor license holder, and regulated by Monetary Authority of Singapore. Please contact CGMSPL at 8 Marina View, 21st Floor Asia Square Tower 1, Singapore 018960, in respect of any matters arising from, or in connection with, the analysis of this document. This Product is intended for recipients who are accredited, expert and institutional investors as defined under the Securities and Futures Act 2001. For Citi Private Bank, the Product is made available in Singapore by Citi Private Bank through Citibank, N.A., Singapore Branch. Citibank N.A., Singapore Branch is a licensed bank in Singapore that is regulated by the Monetary Authority of Singapore. Please contact your Private Banker in Citibank N.A., Singapore Branch if you have any queries on or any matters arising from or in connection with this document. The Product is intended for recipients who are accredited, expert and institutional investors as defined under the Securities and Futures Act 2001. For Citibank Singapore Limited ("CSL"), the Product is distributed in Singapore by CSL to selected Citigold/Citigold Private Clients. CSL provides no independent research or analysis of the substance or in preparation of the Product. Please contact your Citigold/Citigold Private Client Relationship Manager in CSL if you have any queries on or any matters arising from or in connection with this document. The Product is intended for recipients who are accredited investors as defined under the Securities and Futures Act (Cap. 289).

Citigroup Global Markets (Pty) Ltd. is incorporated in the **Republic of South Africa** (company registration number 2000/025866/07) and its registered office is at 145 West Street, Sandton, 2196, Saxonwold. Citigroup Global Markets (Pty) Ltd. is regulated by JSE Securities Exchange South Africa, South African Reserve Bank and the Financial Services Board. The investments and services contained herein are not available to private customers in South Africa.

The Product is made available in the **Republic of China (Taiwan)** through Citigroup Global Markets Taiwan Securities Company Ltd. ("CGMTS"), 14F, 15F and 16F, No. 1, Songzhi Road, Taipei 110, Taiwan, subject to the license scope and the applicable laws and regulations in the Republic of China (Taiwan). CGMTS is regulated by the Securities and Futures Bureau of the Financial Supervisory Commission of Taiwan, the Republic of China (Taiwan). No portion of the Product may be reproduced or quoted in the Republic of China (Taiwan) by the press or any third parties [without the written authorization of CGMTS]. Pursuant to the applicable laws and regulations in the Republic of China (Taiwan), the recipient of the Product shall not take advantage of such Product to involve in any matters in which the recipient may have conflicts of interest. If the Product covers securities which are not allowed to be offered or traded in the Republic of China (Taiwan), neither the Product nor any information contained in the Product shall be considered as advertising the securities or making recommendation of the securities in the Republic of China (Taiwan). The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security or financial products. Any decision to purchase securities or financial products mentioned in the Product must take into account existing public information on such security or the financial products or any registered prospectus.

The Product is made available in **Thailand** through Citicorp Securities (Thailand) Ltd., which is regulated by the Securities and Exchange Commission of Thailand. 399 Interchange 21 Building, 18th Floor, Sukhumvit Road, Klongtoey Nua, Wattana, Bangkok 10110, Thailand.

Disclosure for investors in the Republic of Turkey: The Product is made available in **Turkey** through Citibank AS which is regulated by Capital Markets Board. Tekfen Tower, Eski Buyukdere Caddesi # 209 Kat 2B, 23294 Levent, Istanbul, Turkey. Under Capital Markets Law of Turkey (Law No: 6362), the investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Comments and recommendations stated here rely on the individual opinions of the ones providing these comments and recommendations. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations. Furthermore, Citi Research is a division of Citigroup Global Markets Inc. (the "Firm"), which does and seeks to do business with companies and/or trades on securities covered in this research reports. As a result, investors should be aware that the Firm may have a conflict of interest that could affect the objectivity of this report, however investors should also note that the Firm has in place organisational and administrative arrangements to manage potential conflicts of interest of this nature.

In the **U.A.E.**, these materials (the "Materials") are communicated by Citigroup Global Markets Limited, DIFC branch ("CGML"), an in the Dubai International Financial Center ("DIFC") and licensed and regulated by the Dubai Financial Services Authority ("DFSA", license #CLO221) to Professional Clients and Market Counterparties, as defined in DFSA regulations, only and should not be relied upon or distributed to Retail Clients. Financial products and/or services to which the Materials relate will only be made available to Professional Clients and ties. Citigroup Global Markets Limited DIFC Branch registered address is Level 3, Gate District Building 02, Dubai International Financial Centre and can be contacted on +971 4 509 97 90.

The Product is made available in **United Kingdom** by Citigroup Global Markets Limited, which is authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA. This material may relate to investments or services of a person outside of the UK or to other matters which are not authorised by the PRA nor regulated by the FCA and the PRA and further details as to where this may be the case are available upon request in respect of this material. Citigroup Centre, Canada Square, Canary Wharf, London, E14 5LB.

The Product is made available in **United States** and **Canada** by Citigroup Global Markets Inc., which is a member of FINRA and registered with the US Securities and Exchange Commission. 388 Greenwich Street, New York, NY 10013.

Unless specified to the contrary, within EU Member States, the Product is made available by Citigroup Global Markets Europe AG ("CGME"), which is regulated by the European Central Bank and the German Federal Financial Supervisory Authority (Bundesanstalt für tungsaufsicht-BaFin).

The Product is not to be construed as providing investment services in any jurisdiction where the provision of such services would not be permitted.

Subject to the nature and contents of the Product, the investments described therein are subject to fluctuations in price and/or value and investors may get back less than originally invested. Certain high-volatility investments can be subject to sudden and large falls in value that could equal or exceed the amount invested. The yield and average life of CMOs (collateralized mortgage obligations) referenced in this Product will fluctuate depending on the actual rate at which mortgage holders prepay the mortgages underlying the CMO and changes in current interest rates. Any government agency backing of the CMO applies only to the face value of the CMO and not to any premium paid. Certain investments contained in the Product may have tax implications for private customers whereby levels and basis of taxation may be subject to change. If in doubt, investors should seek advice from a tax adviser. The Product does not purport to identify the nature of the specific market or other risks associated with a particular transaction. Advice in the Product is general and should not be construed as personal advice given it has been prepared without taking account of the objectives, financial situation or needs of any particular investor. Accordingly, investors should, before acting on the advice, consider the appropriateness of the advice, having regard to their objectives, financial situation and needs. Prior to acquiring any financial product, it is the client's responsibility to obtain the relevant offer document for the product and consider it before making a decision as to whether to purchase the product.

Card Insights. Where this report references Card Insights data, Card Insights consists of selected data from a subset of Citi's proprietary credit card transactions. Such data has undergone rigorous security protocols to keep all customer information

confidential and secure; the data is highly aggregated and anonymized so that all unique customer identifiable information is removed from the data prior to receipt by the report's author or distribution to external parties. This data should be considered in the context of other economic indicators and publicly available information. Further, the selected data represents only a subset of Citi's proprietary credit card transactions due to the selection methodology or other limitations and should not be considered as indicative or predictive of the past or future financial performance of Citi or its credit card business.

Citi Research product may source data from dataCentral. dataCentral is a Citi Research proprietary database, which includes the Firm's estimates, data from company reports and feeds from LSEG Data & Analytics. The source for all referenced prices, unless otherwise stated, is DataCentral. Past performance is not a guarantee or reliable indicator of future results. Forecasts are not a guarantee or reliable indicator of future performance. The printed and printable version of the research report may not include all the information <(e.g. certain financial summary information and comparable company data) that is linked to the online version available on the Firm's proprietary electronic distribution platforms.

Where included in this report, MSCI sourced information is the exclusive property of Morgan Stanley Capital International Inc. (MSCI). Without prior written permission of MSCI, this information and any other MSCI intellectual property may not be reproduced, disseminated or used to create any financial products, including any indices. This information is provided on an "as is" basis. The user assumes the entire risk of any use made of this information. MSCI, its affiliates and any third party involved in, or related to, computing or compiling the information hereby expressly disclaim all warranties of originality, accuracy, completeness, merchantability or fitness for a particular purpose with respect to any of this information. Without limiting any of the foregoing, in no event shall MSCI, any of its affiliates or any third party involved in, or related to, computing or compiling the information have any liability for any damages of any kind. MSCI, Morgan Stanley Capital International and the MSCI indexes are services marks of MSCI and its affiliates. Where data is attributed to Morningstar that data is © 2026 Morningstar, Inc. All Rights Reserved. That information: (1) is proprietary to Morningstar and/or its content providers; (2) may not be copied or distributed; and (3) is not warranted to be accurate, complete or timely. Neither Morningstar nor its content providers are responsible for any damages or losses arising from any use of this information.

The Firm accepts no liability whatsoever for the actions of third parties. The Product may provide the addresses of, or contain hyperlinks to, websites. Except to the extent to which the Product refers to website material of the Firm, the Firm has not reviewed the linked site. Equally, except to the extent to which the Product refers to website material of the Firm, the Firm takes no responsibility for, and makes no representations or warranties whatsoever as to, the data and information contained therein. Such address or hyperlink (including addresses or hyperlinks to website material of the Firm) is provided solely for your convenience and information and the content of the linked site does not in any way form part of this document. Accessing such website or following such link through the Product or the website of the Firm shall be at your own risk and the Firm shall have no liability arising out of, or in connection with, any such referenced website.

© 2026 Citigroup Global Markets Inc. Citi Research is a division of Citigroup Global Markets Inc. Citi and Citi and Arc Design are trademarks and service marks of Citigroup Inc. and its affiliates and are used and registered throughout the world. All rights reserved. The research data in this report are not intended to be used for the purpose of (a) determining the price of or amounts due in respect of (or to value) one or more financial products or instruments and/or (b) measuring or comparing the performance of, or defining the asset allocation of a financial product, a portfolio of financial instruments, or a collective investment undertaking, and any such use is strictly prohibited without the prior written consent of Citi Research. Any unauthorized use, duplication, redistribution or disclosure of this report (the "Product"), including, but not limited to, redistribution of the Product by electronic mail, posting of the Product on a website or page, and/or providing to a third party a link to the Product, is prohibited by law and will result in prosecution. The information contained in the Product is intended solely for the recipient and may not be further distributed by the recipient to any third party.

ADDITIONAL INFORMATION IS AVAILABLE UPON REQUEST